

#1 · Professional Communicator and Collaborator

Is medical radiation technology regulated federally or provincially in Canada?

Answer

It is regulated provincially and territorially, not federally. Each province or territory has its own regulatory body, and licensure in one jurisdiction does not automatically transfer to another.

#2 · Professional Communicator and Collaborator

What is CAMRT's role at the national level?

Answer

CAMRT is the national professional association. It develops the national Competency Profile, administers the national certification examination, and sets the national Code of Ethics and Standards of Practice.

#3 · Professional Communicator and Collaborator

What is CMRITO and what does it do?

Answer

CMRITO is the College of Medical Radiation and Imaging Technologists of Ontario, the provincial regulatory college for MRTs in Ontario. It issues the legal certificate of registration required to practise, separate from CAMRT's national association role.

#4 · Professional Communicator and Collaborator

What is the key difference between a professional association and a regulatory college?

Answer

A professional association advances the profession through education, advocacy, and standards, while a regulatory college exists for public protection, with statutory authority to register, investigate, discipline, and revoke a practitioner's certificate.

#5 · Professional Communicator and Collaborator

Fill in the blank: A technologist moving to a new province ____ automatically retain their registration and must apply for registration in the new jurisdiction.

Answer

does not; reciprocity exists but is not automatic, and an application process is required in the new province.

#6 · Professional Communicator and Collaborator

What is scope of practice defined by, for a medical radiation technologist?

Answer

Three overlapping sources: provincial or territorial legislation, the employing institution's policies and protocols, and the individual technologist's own demonstrated competence.

#7 · Professional Communicator and Collaborator

A radiologist asks a technologist to interpret an image and tell the patient the diagnosis. What should the technologist do?

Answer

Decline, since interpretation and diagnosis fall outside an MRT's scope of practice. The technologist may share factual, non-diagnostic information but must direct diagnostic questions to the radiologist or referring physician.

#8 · Professional Communicator and Collaborator

What is a medical directive?

Answer

A standing order from a physician or radiologist that permits a technologist to perform a specific delegated act under specific conditions, without requiring a direct individual order each time.

#9 · Professional Communicator and Collaborator

If a technologist performs a delegated act without proper authorization, who is accountable?

Answer

The technologist remains personally accountable under most provincial regulatory frameworks, even if a physician requested the task.

#10 · Professional Communicator and Collaborator

What is the correct decision sequence when a technologist is unsure whether a task is within their scope of practice?

Answer

First confirm it is permitted under provincial legislation and college standards, second confirm employer policy and any medical directive authorize it, and third honestly assess personal competence. Proceed only if all three are satisfied; otherwise decline and escalate.

#11 · Professional Communicator and Collaborator

What does the beneficence and non-maleficence pillar of the CAMRT Code of Ethics require in imaging practice?

Answer

Acting in the patient's best interest and avoiding harm, which underlies the ALARA framework for radiation dose, infection control, safe patient handling, and catching errors such as wrong-patient situations before they cause harm.

#12 · Professional Communicator and Collaborator

What does ALARA stand for?

Answer

As low as reasonably achievable, the framework guiding minimization of radiation dose to patients and staff.

#13 · Professional Communicator and Collaborator

What does the accountability pillar of the Code of Ethics require?

Answer

That a technologist accepts responsibility for their own professional decisions and actions, including the outcomes of delegated tasks they choose to perform.

#14 · Professional Communicator and Collaborator

What does the justice and equity pillar of the CAMRT Code of Ethics require?

Answer

Fair, unbiased treatment of all patients without discrimination based on race, ethnicity, religion, gender identity, sexual orientation, disability, socioeconomic status, or any other personal characteristic.

#15 · Professional Communicator and Collaborator

Is professional competence a one-time achievement or an ongoing obligation under the CAMRT Code of Ethics?

Answer

An ongoing ethical duty. A technologist must practise only within the bounds of their actual current competence, pursue continuing education, and recognize when a situation exceeds their skill.

#16 · Professional Communicator and Collaborator

What constitutes implied consent for a routine diagnostic radiography examination?

Answer

The patient's cooperative presence in the examination room and voluntary positioning for the requested view. This does not authorize adding views, contrast media, or any procedure with materially greater risk.

#17 · Professional Communicator and Collaborator

A competent patient withdraws consent partway through an examination. What must the technologist do?

Answer

Stop the examination promptly, document the withdrawal, and notify the appropriate physician. A patient may withdraw consent at any point, even mid-procedure.

#18 · Professional Communicator and Collaborator

Is capacity to consent a fixed, global trait or is it decision-specific?

Answer

Capacity is decision-specific and time-specific. A patient may lack capacity for one type of decision while retaining it for another, and capacity can fluctuate, for example while recovering from sedation.

#19 · Professional Communicator and Collaborator

What happens when a patient lacks capacity to consent and no emergency exists?

Answer

Consent must be obtained from a substitute decision-maker in accordance with provincial legislation, which typically establishes a hierarchy such as spouse, adult child, parent, or sibling.

#20 · Professional Communicator and Collaborator

What is PHIPA?

Answer

The Personal Health Information Protection Act, Ontario's statute governing the collection, use, and disclosure of personal health information by health information custodians and providers.

#21 · Professional Communicator and Collaborator

What is the "circle of care" or "need to know" principle?

Answer

Access to a patient's information in systems such as PACS or RIS should be limited to what is genuinely needed for the technologist's current task, never browsed out of curiosity even when technically possible.

#22 · Professional Communicator and Collaborator

A technologist wants to post about an unusual case on social media without naming the patient. Is this acceptable?

Answer

No. Posting any identifiable patient information or details without explicit consent is a serious breach of confidentiality, even without naming the patient, because small details can allow identification.

#23 · Professional Communicator and Collaborator

A technologist discovers a privacy breach caused by a colleague. What is the professional obligation?

Answer

To report it through the facility's incident reporting process, regardless of who caused the breach or whether it was accidental or intentional.

#24 · Professional Communicator and Collaborator

What is the correct way to fix an error discovered in a patient record after the fact?

Answer

Make a clearly labelled amendment or addendum. Never delete or overwrite the original entry in a way that conceals what was originally recorded.

#25 · Professional Communicator and Collaborator

What qualities should good documentation have?

Answer

Contemporaneous, factual and objective rather than speculative, complete, and legible or correctly entered with a proper amendment trail if electronic.

#26 · Professional Communicator and Collaborator

What is a "just culture" approach to incident reporting?

Answer

An approach that distinguishes honest mistakes, which should be reported and addressed through systemic improvement, from at-risk behaviour addressed through coaching, and from reckless conduct that knowingly disregards a substantial risk, which warrants discipline.

#27 · Professional Communicator and Collaborator

Does the obligation to report an incident depend on whether the technologist personally caused it?

Answer

No. The obligation to report exists whether the technologist caused the incident, was merely present, or discovered it after the fact. Concealing a known incident is itself a serious breach.

#28 · Professional Communicator and Collaborator

What is the SBAR framework used for?

Answer

Situation, background, assessment, and recommendation, a structured tool used for handoff communication to ensure critical patient information is not lost during transitions of care.

#29 · Professional Communicator and Collaborator

A technologist notices a possible pneumothorax on a portable chest radiograph. What should the technologist do?

Answer

Promptly notify the radiologist or appropriate physician following the facility's critical results reporting policy, rather than completing the study and moving on without flagging the finding.

#30 · Professional Communicator and Collaborator

A patient asks the technologist directly what their image shows. What is the professionally correct response?

Answer

Acknowledge the question with empathy while clearly and kindly redirecting the diagnostic question to the appropriate physician, without appearing dismissive or evasive.

#31 · Professional Communicator and Collaborator

What is the difference between cultural awareness and cultural safety?

Answer

Cultural awareness recognizes that cultural differences exist, while cultural safety requires the provider to critically examine their own position, power, and biases, and lets the patient define whether the interaction felt safe and respectful.

#32 · Professional Communicator and Collaborator

An Indigenous patient asks for a family member or Elder to be present during an examination. What is the culturally safe response?

Answer

Respect the request where facility policy and clinical circumstances allow it, as part of building trust and responding to historical and ongoing experiences of discrimination in health care.

#33 · Professional Communicator and Collaborator

A patient's family member offers to translate sensitive medical information. What should the technologist do instead?

Answer

Use professional interpretation services rather than relying on an accompanying family member to translate, since language barriers can exist even for patients who speak English or French with unfamiliar clinical terms.

#34 · Professional Communicator and Collaborator

Who is responsible for tracking and meeting continuing competency requirements?

Answer

The technologist personally, not the employer, although many employers support continuing education through funding or protected time.

#35 · Professional Communicator and Collaborator

A technologist believes an ordered examination is clinically inappropriate given the patient's presentation. What is the professionally correct action?

Answer

Raise the concern through appropriate channels, such as clarifying with the referring physician, rather than simply proceeding against their own professional judgment or refusing without explanation.

#36 · Care Provider

What three points in the imaging encounter require a structured patient assessment?

Answer

Pre-procedure, during the procedure, and post-procedure. Each stage has its own focus, from screening for contraindications before the exam to confirming baseline status has returned before discharge.

#37 · Care Provider

During the pre-procedure assessment, what should the technologist screen for besides confirming identity?

Answer

Contraindications such as pregnancy or allergy to contrast media, along with the patient's general condition (gait, colour, level of consciousness, and apparent distress).

#38 · Care Provider

Why are verbal check-ins between exposures important during a procedure?

Answer

They are a monitoring tool, not just a courtesy. A patient who stops answering appropriately, becomes unusually quiet, or reports new chest pain or shortness of breath needs immediate reassessment before the exam continues.

#39 · Care Provider

Scenario: A patient has just had an intravenous contrast injection. What discharge instruction should the technologist communicate during the post-procedure assessment?

Answer

Increased fluid intake after intravenous contrast, along with confirming the patient can safely stand and walk or that appropriate assistance has been arranged.

#40 · Care Provider

Normal resting heart rate range for an average healthy adult: ____ to ____ beats per minute.

Answer

60 to 100 beats per minute.

#41 · Care Provider

Normal resting respiratory rate range for an average healthy adult.

Answer

12 to 20 breaths per minute.

#42 · Care Provider

What oxygen saturation on room air is considered normal for an adult at rest?

Answer

95 percent or higher.

#43 · Care Provider

How does an infant's normal resting heart rate compare to an adult's?

Answer

It runs faster, normally between 100 and 160 beats per minute, decreasing gradually toward adult ranges through childhood.

#44 · Care Provider

At what systolic blood pressure is a patient considered hypotensive and in need of reassessment?

Answer

Below 90 mmHg.

#45 · Care Provider

Scenario: A resting adult patient has a heart rate of 110 beats per minute with no other symptoms. What vital sign finding is this, and what should the technologist do?

Answer

This is tachycardia (above 100 bpm at rest). The technologist should escalate to the appropriate regulated staff, such as a radiologist, nurse, or physician, rather than diagnose it independently.

#46 · Care Provider

An oxygen saturation reading below ____ percent warrants prompt reassessment and possible activation of the emergency response protocol.

Answer

92 percent.

#47 · Care Provider

Do standard (routine) precautions apply only to patients with a known infection?

Answer

No. Standard precautions apply to every patient interaction regardless of known infection status, since a patient's infectious status is frequently unknown at the point of care.

#48 · Care Provider

List the core elements of standard precautions.

Answer

Hand hygiene before and after every patient contact, gloves when contact with blood or body fluids is anticipated, safe sharps handling and disposal, routine cleaning and disinfection of equipment, and respiratory hygiene and cough etiquette.

#49 · Care Provider

What are the "4 Moments for Hand Hygiene" adapted for imaging practice?

Answer

Before touching a patient, before a clean or aseptic procedure, after a body fluid exposure risk, and after touching a patient or their immediate environment.

#50 · Care Provider

When should soap and water be used for hand hygiene instead of alcohol-based hand rub?

Answer

When hands are visibly contaminated, or when caring for a patient with *Clostridioides difficile*, since alcohol-based products do not reliably kill *C. difficile* spores.

#51 · Care Provider

Which transmission-based precaution category applies to MRSA, VRE, and C. difficile?

Answer

Contact precautions, requiring gloves and a gown for any contact with the patient or their environment, plus dedicated or disinfected equipment.

#52 · Care Provider

Approximately how far can large respiratory droplets travel, defining the distance at which droplet precautions apply?

Answer

Roughly one to two metres.

#53 · Care Provider

Scenario: A patient with confirmed tuberculosis needs a chest X-ray. What PPE and room considerations apply?

Answer

Airborne precautions apply: a fit-tested N95 respirator or equivalent, and ideally transport and imaging in a negative-pressure room or at the end of the day's schedule to minimize exposure of subsequent patients.

#54 · Care Provider

Name three pathogens that require droplet precautions.

Answer

Influenza, pertussis, and some strains of meningococcal disease.

#55 · Care Provider

What is the correct order for donning PPE?

Answer

Hand hygiene, gown, mask or respirator, eye protection or face shield, and finally gloves.

#56 · Care Provider

Which PPE item is removed first when doffing, and which is removed last?

Answer

Gloves are removed first because they carry the heaviest contamination. The mask or respirator is removed last, touched only by its ties or ear loops.

#57 · Care Provider

What is the full doffing sequence after gloves are removed?

Answer

Hand hygiene, then eye protection, then gown, then the mask or respirator, followed by hand hygiene again immediately after doffing is complete.

#58 · Care Provider

Why is the front of a used mask considered contaminated?

Answer

It has been in contact with respiratory droplets or aerosols filtered by the mask, so it should only be handled by its ties or ear loops during removal, and touching it with a bare hand requires immediate hand hygiene.

#59 · Care Provider

What contact surfaces of fixed imaging equipment require disinfection between patients?

Answer

The table, the tube housing handles, and positioning devices, along with any portable equipment or cassettes brought into an isolation room.

#60 · Care Provider

What is the core principle of proper body mechanics during a manual patient transfer?

Answer

A wide, stable base of support, bending at the knees rather than the waist, keeping the load close to the body, and avoiding twisting the spine while lifting.

#61 · Care Provider

When should mechanical lift devices or slide boards be used for a patient transfer?

Answer

Whenever a patient cannot bear their own weight or actively assist, and a second person should always be summoned for any transfer involving a dependent or heavy patient.

#62 · Care Provider

Why are elderly patients at higher risk of fracture during transfers?

Answer

They often have decreased bone density, making them vulnerable to fracture from even a minor fall or an overly forceful transfer.

#63 · Care Provider

What is orthostatic hypotension, and how should position changes for an elderly patient account for it?

Answer

A drop in blood pressure when moving from lying to sitting to standing. Position changes should occur slowly and in stages, with a pause at each stage to reassess for dizziness.

#64 · Care Provider

What must a technologist confirm before positioning a bariatric patient on the imaging table?

Answer

The imaging table's weight capacity, rather than assuming standard equipment is adequate; weight-rated equipment and wider tables or extension devices are typically required.

#65 · Care Provider

Scenario: A trauma patient with unconfirmed spinal clearance needs to be repositioned. What technique is required, and with how many people?

Answer

Log-roll technique, using a minimum of three to four people to maintain spinal alignment. The technologist must never remove a cervical collar or spinal immobilization device without explicit physician clearance.

#66 · Care Provider

How can a parent or caregiver's presence help during a pediatric patient's imaging exam?

Answer

It can reduce anxiety and improve cooperation. With appropriate shielding, and provided they are not pregnant, there is no restriction against their radiation exposure beyond standard precautions.

#67 · Care Provider

What positioning practice protects skin integrity during a long CT protocol or fluoroscopic procedure?

Answer

Padding at bony prominences, minimizing total table time where possible, and checking skin condition on patients who are unconscious, sedated, or otherwise unable to report discomfort.

#68 · Care Provider

What is the purpose of an immobilization device in imaging?

Answer

To reduce voluntary and involuntary motion to preserve image quality while protecting the patient from injury caused by movement during exposure.

#69 · Care Provider

Name four common types of immobilization devices used in imaging.

Answer

Compression bands, sponge and foam positioning aids, sandbags, and Velcro straps (specialized pediatric immobilizers are also used).

#70 · Care Provider

What risk does an improperly applied immobilization device create?

Answer

It can compromise circulation or respiration, so any device must be monitored throughout the examination and applied only as tightly as needed to achieve the clinical goal.

#71 · Care Provider

What is required before chemical or physical restraint may be used purely for behavioural control rather than to protect the patient during a procedure?

Answer

Generally a physician order and informed consent or substitute decision-maker consent, since this use raises significant legal and ethical concerns.

#72 · Care Provider

Why are diagnostic imaging departments a common site for patient falls?

Answer

Because of frequent transitions between wheelchairs, stretchers, and imaging tables, often on unfamiliar equipment and sometimes with the patient's clothing altered or vision reduced by removed glasses.

#73 · Care Provider

List four fall prevention strategies used in a diagnostic imaging department.

Answer

Standardized fall-risk screening on arrival, never leaving an at-risk patient unattended, keeping side rails raised and stretchers or beds locked, and lowering the imaging table to its minimum height during transfers.

#74 · Care Provider

Scenario: A technologist must briefly step into the control booth, leaving an ambulatory but unsteady patient on the table. What should be provided before leaving?

Answer

A call bell and clear verbal instructions, such as "do not get up without help."

#75 · Care Provider

What is syncope, and what commonly triggers it in the imaging setting?

Answer

A transient loss of consciousness caused by a temporary reduction in cerebral blood flow. It is often triggered by fear, pain, prolonged standing, or a vasovagal response to venipuncture.

#76 · Care Provider

Name three presyncope warning signs.

Answer

Pallor, diaphoresis (sweating), and lightheadedness, along with reports of "feeling faint."

#77 · Care Provider

Scenario: A standing patient reports feeling lightheaded and sweaty. What should the technologist do immediately?

Answer

Assist the patient to a safe seated or supine position, elevate the legs if supine, loosen restrictive clothing, and monitor vital signs until the episode resolves. The patient should never be left standing.

#78 · Care Provider

What is the correct technologist response to a patient having a seizure in the imaging department?

Answer

Protect the patient from injury rather than restrain the movements: ease them to the floor or table, clear the area of hard or sharp objects, cushion the head, and once movements permit, turn them onto their side to protect the airway.

#79 · Care Provider

Should an object ever be placed in a seizing patient's mouth?

Answer

No, never. This is a common misconception; nothing should be placed in the mouth during a seizure.

#80 · Care Provider

At what point does a seizure become status epilepticus, requiring emergency assistance?

Answer

When it lasts longer than five minutes, or when a second seizure begins before the patient recovers consciousness from the first.

#81 · Care Provider

What are the early signs of anaphylaxis to contrast media?

Answer

Urticaria (hives), pruritus, flushing, and a feeling of impending doom, which can progress rapidly to angioedema, bronchospasm, hypotension, and cardiovascular collapse.

#82 · Care Provider

Contrast reactions are classically graded into three severity tiers. Name them.

Answer

Mild, moderate, and severe.

#83 · Care Provider

Scenario: A patient develops limited urticaria and a sensation of warmth after contrast injection, with no other symptoms. What grade of reaction is this, and what response is typically needed?

Answer

A mild reaction. It typically requires only observation, reassurance, and monitoring, since these reactions are usually self-limited.

#84 · Care Provider

What findings characterize a moderate contrast reaction?

Answer

More extensive urticaria, facial or laryngeal edema without severe respiratory compromise, mild bronchospasm, or a transient drop in blood pressure with symptoms.

#85 · Care Provider

What response is required for a severe contrast reaction?

Answer

Immediate activation of the emergency response system, administration of epinephrine per protocol or physician order, high-flow oxygen, and initiation of basic life support if the patient loses pulse or respiration.

#86 · Care Provider

What must every diagnostic imaging department administering intravenous contrast have in place?

Answer

An emergency response protocol, a stocked emergency cart with epinephrine, and staff trained to recognize and respond to reactions immediately.

#87 · Care Provider

Canadian adult basic life support follows the ____ - ____ - ____ sequence.

Answer

C-A-B: Compressions first, then Airway, then Breathing.

#88 · Care Provider

What is the recommended chest compression rate and depth for an adult during CPR?

Answer

100 to 120 compressions per minute, at a depth of approximately 5 centimetres, allowing full chest recoil between compressions.

#89 · Care Provider

What is the standard compression-to-breath ratio for a lone rescuer performing adult CPR?

Answer

30 compressions to 2 rescue breaths.

#90 · Care Provider

Scenario: An AED has just delivered a shock to a patient in cardiac arrest. What should happen next?

Answer

Compressions should resume immediately after the shock is delivered, or after the device advises no shock is indicated.

#91 · Care Provider

Why are technologists generally required to maintain current BLS certification?

Answer

Because the imaging environment places them at genuine risk of encountering a cardiac or respiratory arrest, given their direct patient contact and use of contrast media.

#92 · Care Provider

What is the appropriate intervention for a conscious adult who is choking and cannot cough, speak, or breathe?

Answer

Abdominal thrusts (the Heimlich manoeuvre), until the obstruction clears or the patient becomes unresponsive, at which point CPR should begin.

#93 · Care Provider

What is the minimum standard for verifying patient identity before any examination begins?

Answer

At least two independent identifiers, such as full name and date of birth, or full name and a unique medical record or health card number, verified against the requisition.

#94 · Care Provider

Why is a verbal confirmation like "Are you Mr. Smith?" insufficient as a sole patient identifier?

Answer

A confused, hard-of-hearing, or agreeable patient may answer affirmatively to any name offered, so it cannot be relied on alone.

#95 · Care Provider

How should identity be confirmed for a patient who cannot verbally respond, such as an unconscious trauma patient?

Answer

By relying on the identification wristband, accompanying staff or family member confirmation, and cross-referencing with the requisition and any accompanying chart.

#96 · Care Provider

At what additional point, besides the start of the exam, should the two-identifier standard be repeated?

Answer

Before contrast administration and before any invasive component of a procedure.

#97 · Care Provider

Who must be screened for possible pregnancy before an examination involving ionizing radiation to the pelvis or abdomen?

Answer

Every person of childbearing potential, unless the examination is emergent and cannot be safely delayed.

#98 · Care Provider

What is the "10-day rule" in radiation protection?

Answer

A historical guideline recommending that elective abdominal and pelvic radiographic exams on a person of childbearing potential be scheduled within the first 10 days following the onset of menstruation, the period considered least likely to coincide with an early, unrecognized pregnancy.

#99 · Care Provider

Has the strict 10-day rule been fully replaced in contemporary practice?

Answer

It has largely been relaxed in favour of a pregnancy questionnaire at the time of the visit, since a missed period is a stronger practical indicator than a rigid calendar window, though technologists must still know their own department's current policy.

#100 · Care Provider

Scenario: A patient's pregnancy cannot be ruled out and the requested exam is not emergent. What should happen next?

Answer

The referring physician and, where appropriate, a radiologist should be consulted to weigh the clinical necessity of the exam against fetal radiation dose, and non-ionizing alternatives such as ultrasound or MRI should be considered.

#101 • Care Provider

Why are isolation patients ideally scheduled at the end of the day's imaging list?

Answer

To reduce the number of subsequent patients who might be exposed to a contaminated room; alternatively, a dedicated portable unit may be brought to the patient's room instead.

#102 • Care Provider

What must happen to porous equipment, such as positioning sponges, that entered an isolation environment and cannot be adequately disinfected?

Answer

It must be discarded per department policy before reuse on another patient.

#103 · Care Provider

Where should a urinary catheter drainage bag generally be kept relative to the bladder, and why?

Answer

Below bladder level, to prevent backflow of urine into the bladder.

#104 · Care Provider

What is the risk of clamping a chest tube without a specific order?

Answer

Tension pneumothorax; a chest tube drainage unit must be kept upright and below chest level and should never be clamped without explicit orders.

#105 · Care Provider

What should a technologist do before transferring or repositioning a patient with multiple tubes and lines in place?

Answer

Identify every tube and line, note where it enters the body and where it connects, and ensure adequate slack so movement does not create tension on the site.

#106 · Care Provider

Scenario: A technologist notices new bleeding at a wound site during a patient transfer. What is the correct action?

Answer

Report it immediately to nursing or the treating team rather than managing it independently.

#107 · Care Provider

Name three strategies to reduce claustrophobic anxiety during an MRI examination.

Answer

Explaining the procedure and its duration clearly in advance, offering a blindfold, music, or a mirror device, and providing a call bell or squeeze bulb the patient can use to stop the exam if needed.

#108 · Care Provider

When communicating with a pediatric patient, what type of language should be avoided?

Answer

Words that could be misinterpreted as frightening, such as describing a procedure as "shooting" an X-ray. Simple, honest, age-appropriate language should be used instead.

#109 • Care Provider

What communication adjustments help a geriatric patient with possible hearing loss or cognitive change?

Answer

Speaking clearly and at a moderate pace, facing the patient directly to support lip-reading, allowing extra time to process instructions and reposition, and including an accompanying caregiver in explanations when needed.

#110 • Care Provider

What communication practice, used across all age groups, builds the trust that produces a safer and more diagnostically successful examination?

Answer

Using the patient's name, explaining what will happen and why before every step, and checking for understanding rather than simply issuing instructions.

#111 · Safe Work Principles and Radiation Safety

What are deterministic effects of radiation?

Answer

Tissue reactions that occur only above a threshold dose, with severity increasing as dose increases above that threshold. Skin erythema, epilation, and cataract formation are classic examples.

#112 · Safe Work Principles and Radiation Safety

What are stochastic effects of radiation?

Answer

Effects with no dose threshold, where the probability of occurrence rises with dose but the severity of the effect does not. Cancer induction and heritable genetic effects are the two of greatest concern in diagnostic radiography.

#113 · Safe Work Principles and Radiation Safety

Fill in the blank: Below the threshold dose, a _____ effect will not occur at all.

Answer

Deterministic.

#114 · Safe Work Principles and Radiation Safety

What model underlies radiation protection philosophy given that stochastic effects have no threshold?

Answer

The linear no-threshold model, which assumes any dose carries some small risk, however small that risk may be at diagnostic dose levels.

#115 · Safe Work Principles and Radiation Safety

What does ALARA stand for and what does it mean in practice?

Answer

As Low As Reasonably Achievable. It means technologists always work to minimize dose to patients, staff, and the public, taking economic and social factors into account, rather than aiming for a fixed numeric dose target.

#116 · Safe Work Principles and Radiation Safety

What three practical tools support the ALARA principle?

Answer

Minimizing time near a radiation source, maximizing distance from the source, and using shielding wherever practical.

#117 · Safe Work Principles and Radiation Safety

Scenario: A technologist performs a long fluoroscopically guided procedure with extended beam-on time, and the patient later develops a skin injury at the entrance site. Which category of radiation effect does this represent?

Answer

A deterministic effect. Prolonged beam-on time in fluoroscopy can exceed the threshold dose for skin injury, and the injury severity would increase with further dose above that threshold.

#118 · Safe Work Principles and Radiation Safety

What is the gray (Gy) the SI unit of?

Answer

Absorbed dose, the amount of energy deposited per unit mass of tissue, specifically one joule of energy absorbed per kilogram of matter.

#119 · Safe Work Principles and Radiation Safety

What is the sievert (Sv) the SI unit of?

Answer

Equivalent dose and effective dose, quantities that incorporate weighting factors to reflect the biological risk of radiation exposure.

#120 · Safe Work Principles and Radiation Safety

Why are equivalent dose in sieverts and absorbed dose in grays numerically similar for diagnostic x-ray work?

Answer

Because the radiation weighting factor for x-ray and gamma radiation is one, even though the two quantities remain conceptually distinct.

#121 · Safe Work Principles and Radiation Safety

What is the becquerel (Bq) the SI unit of, and what does it NOT measure?

Answer

It is the SI unit of radioactivity, one radioactive disintegration per second. It measures the rate of decay of a source, not the dose received by a person.

#122 · Safe Work Principles and Radiation Safety

Fill in the blank: In diagnostic radiography, doses are commonly expressed in _____ and occupational limits in _____.

Answer

Milligray (mGy); millisievert (mSv).

#123 · Safe Work Principles and Radiation Safety

Which SI units does the CAMRT exam expect familiarity with, as opposed to legacy units like the roentgen, rad, and rem?

Answer

The gray, sievert, and becquerel, since Canadian practice today is based on SI units.

#124 · Safe Work Principles and Radiation Safety

What is the commonly cited occupational effective dose limit in Canada for radiation workers?

Answer

50 millisievert in any single year, with a cumulative limit of 100 millisievert over any five-year period.

#125 • Safe Work Principles and Radiation Safety

Why does the occupational dose framework average exposure over a five-year period rather than only capping a single year?

Answer

It recognizes that occupational exposure should be managed over time, preventing any single year from reaching an excessive dose while allowing normal year-to-year variation.

#126 • Safe Work Principles and Radiation Safety

Are occupational dose limits meant to be treated as a target for technologists to approach?

Answer

No. They are an upper regulatory boundary, not a target. ALARA drives daily practice to keep actual doses as far below these limits as reasonably achievable.

#127 · Safe Work Principles and Radiation Safety

How do dose limits for the lens of the eye and for members of the general public compare to the whole-body occupational limit?

Answer

Both are lower. The lens of the eye has a separate, lower limit due to its sensitivity to cataract formation, and members of the public are held to a substantially lower limit than occupationally exposed workers.

#128 · Safe Work Principles and Radiation Safety

What are the two dosimeter technologies that dominate Canadian practice?

Answer

Thermoluminescent dosimeters (TLDs) and optically stimulated luminescence (OSL) dosimeters.

#129 · Safe Work Principles and Radiation Safety

How does a TLD work?

Answer

A crystalline material, commonly lithium fluoride, absorbs radiation energy and stores it as trapped electrons. When later heated in a laboratory reader, the trapped electrons release energy as light proportional to the absorbed dose.

#130 · Safe Work Principles and Radiation Safety

How does an OSL dosimeter differ from a TLD in its readout mechanism?

Answer

An OSL dosimeter, often made of aluminum oxide, is read using a laser or LED light source rather than heat, stimulating release of the stored energy as measurable light.

#131 · Safe Work Principles and Radiation Safety

Why have OSL dosimeters become widely adopted?

Answer

They allow reanalysis of the same dosimeter, offer good sensitivity at low doses, and provide reliable long-term stability of the recorded dose.

#132 · Safe Work Principles and Radiation Safety

Where should a personal dosimeter be worn, and why?

Answer

At collar level, outside any lead apron, facing the primary beam or the most likely direction of scatter, to estimate dose to the unshielded head, neck, and thyroid.

#133 • Safe Work Principles and Radiation Safety

Why might a facility issue a second dosimeter under the apron at waist level?

Answer

For staff regularly involved in fluoroscopy, allowing calculation of a more accurate effective dose using an algorithm that combines both dosimeter readings.

#134 • Safe Work Principles and Radiation Safety

Scenario: A technologist leaves her dosimeter badge in a hot car over the weekend instead of wearing it. What is the consequence?

Answer

The dosimeter may produce a false dose record that misrepresents her actual exposure, undermining the purpose of the monitoring program. Dosimeters must never be left where they could receive spurious exposure.

#135 · Safe Work Principles and Radiation Safety

What should happen if a dosimeter reading appears unexpectedly high?

Answer

The facility's radiation safety officer investigates promptly, since the reading may reflect a genuine overexposure requiring corrective action or an artifact such as the dosimeter being left near a source unworn.

#136 · Safe Work Principles and Radiation Safety

What are the three cardinal principles of radiation protection?

Answer

Time, distance, and shielding.

#137 · Safe Work Principles and Radiation Safety

How does the inverse square law relate distance from a source to radiation intensity?

Answer

The intensity of radiation from a point source is inversely proportional to the square of the distance from that source, so doubling distance reduces intensity to one quarter.

#138 · Safe Work Principles and Radiation Safety

Fill in the blank: Tripling the distance from a radiation source reduces intensity to _____ of its original value.

Answer

One ninth.

#139 · Safe Work Principles and Radiation Safety

Name two ways of reducing time near a radiation source during fluoroscopy.

Answer

Using pulsed fluoroscopy modes rather than continuous fluoroscopy, and using last-image-hold features instead of re-exposing to review an image.

#140 · Safe Work Principles and Radiation Safety

Why is lead the primary material used for shielding in diagnostic radiography?

Answer

Because of its high atomic number and density, which make it highly effective at attenuating x-ray photons across the diagnostic energy range.

#141 · Safe Work Principles and Radiation Safety

How is lead apron protection typically rated?

Answer

By lead equivalence, commonly expressed in millimetres of lead equivalent, with 0.25 mm, 0.35 mm, and 0.5 mm being common ratings.

#142 · Safe Work Principles and Radiation Safety

What is the tradeoff facilities weigh when selecting lead apron thickness?

Answer

A higher lead equivalence provides greater attenuation but adds weight and can contribute to musculoskeletal strain, so apron thickness is chosen to balance protection against practical wearability.

#143 • Safe Work Principles and Radiation Safety

Who should wear a wraparound apron, and why?

Answer

Staff who may have their back toward the radiation source during a procedure, such as when positioning a patient during fluoroscopy, since a wraparound apron gives continuous coverage front and back.

#144 • Safe Work Principles and Radiation Safety

Why is a thyroid collar considered standard protective equipment for anyone remaining in the room during fluoroscopy?

Answer

The thyroid is a radiosensitive organ that sits outside the coverage area of a standard apron, so it needs its own separate shielding from scattered radiation.

#145 · Safe Work Principles and Radiation Safety

What is a key limitation of lead-lined gloves that technologists must remember?

Answer

They offer only limited attenuation compared to the intensity of the primary beam and must never be relied on to justify placing the hands directly in the primary, unattenuated beam.

#146 · Safe Work Principles and Radiation Safety

How has current radiation protection practice regarding patient gonadal shielding changed from traditional default use?

Answer

Practice has moved toward a more selective, evidence-based approach, recognizing that modern automatic exposure control, collimation, and technique selection already reduce dose substantially, and that improperly placed shielding can obscure anatomy or increase dose.

#147 · Safe Work Principles and Radiation Safety

What is the primary purpose of collimation in patient dose reduction?

Answer

Restricting the x-ray beam to the smallest field size necessary reduces the volume of tissue irradiated and improves image quality by reducing scattered radiation reaching the image receptor.

#148 · Safe Work Principles and Radiation Safety

How can exposure factor selection reduce patient dose while maintaining image quality?

Answer

Higher kVp with correspondingly lower mAs can often maintain diagnostic quality while reducing dose, since kVp has a stronger relationship with radiation quantity than mAs alone.

#149 · Safe Work Principles and Radiation Safety

What is the tradeoff involved in using a grid during radiography?

Answer

Grids absorb scattered radiation and improve contrast for thicker body parts, but they require increased mAs to compensate, which increases patient dose, so they should not be used reflexively on thin or pediatric patients.

#150 · Safe Work Principles and Radiation Safety

Why is avoiding repeat exposures one of the most effective dose reduction strategies?

Answer

A repeated exposure delivers dose to the patient with no corresponding diagnostic benefit from the original, discarded image, so preventing repeats through good technique eliminates entirely unnecessary dose.

#151 • Safe Work Principles and Radiation Safety

What is the purpose of an automatic exposure control (AEC) system?

Answer

It terminates exposure once sufficient radiation has reached the image receptor for the selected technique, helping avoid overexposure from an excessively long manual exposure time.

#152 • Safe Work Principles and Radiation Safety

What does declaring a pregnancy trigger for an occupationally exposed technologist in Canadian practice?

Answer

A separate, more conservative dose limit intended to protect the developing embryo or fetus for the remainder of the declared pregnancy, along with additional monitoring and possible duty adjustments.

#153 • Safe Work Principles and Radiation Safety

Is declaring a pregnancy mandatory for a technologist in Canadian practice?

Answer

No, it is voluntary. However, declaring it allows the facility to apply additional monitoring and, where appropriate, adjust work duties to help protect fetal dose.

#154 • Safe Work Principles and Radiation Safety

Scenario: A technologist declares her pregnancy. What change is commonly made to monitor her exposure?

Answer

A second dosimeter is commonly worn at waist level under the apron to more directly estimate fetal dose once the pregnancy is declared.

#155 · Safe Work Principles and Radiation Safety

Why is an urgently needed diagnostic examination rarely contraindicated outright by patient pregnancy alone?

Answer

Because the radiation risk from a properly performed diagnostic examination is generally small relative to the risk of a missed or delayed diagnosis, so clinical necessity is weighed against fetal dose rather than automatically refusing the examination.

#156 · Safe Work Principles and Radiation Safety

How can technique be optimized when imaging a pregnant patient whose examination is clinically justified and cannot be deferred?

Answer

Through tight collimation, appropriate exposure factor selection, and abdominal shielding where practical and appropriate to the examination, while still obtaining the diagnostic information needed.

#157 · Safe Work Principles and Radiation Safety

What three factors are considered when calculating structural shielding for an imaging room?

Answer

Workload (anticipated volume and type of examinations), use factor (fraction of time the beam points toward a particular wall), and occupancy factor (how frequently and for how long people occupy the adjacent space).

#158 · Safe Work Principles and Radiation Safety

What is the difference between a primary barrier and a secondary barrier?

Answer

A primary barrier must attenuate the direct, unattenuated primary beam and requires substantially more lead equivalence, while a secondary barrier need only attenuate weaker leakage and scattered radiation from walls the beam never directly strikes.

#159 • Safe Work Principles and Radiation Safety

What is the purpose of a radiation protection survey performed after installing new or modified radiographic equipment?

Answer

A qualified expert uses calibrated survey instruments to measure radiation levels outside the shielded barriers, confirming that the shielding design performs as intended and that levels in occupied adjacent areas remain within acceptable limits.

#160 • Safe Work Principles and Radiation Safety

What is the difference between quality control (QC) and quality assurance (QA) for x-ray equipment?

Answer

Quality control refers to the specific technical tests performed on equipment, while quality assurance describes the broader administrative and organizational program that ensures those tests are performed, documented, and acted upon.

#161 · Safe Work Principles and Radiation Safety

What does a kVp accuracy and reproducibility test check?

Answer

That the actual kilovoltage produced by the tube matches the selected value and remains consistent across repeated exposures at the same setting.

#162 · Safe Work Principles and Radiation Safety

What does a linearity test confirm for x-ray equipment?

Answer

That changing mAs through different combinations of mA and time produces a proportional change in output.

#163 · Safe Work Principles and Radiation Safety

What does half-value layer measurement assess, and why does it matter?

Answer

It assesses the effective filtration of the x-ray beam, and it relates directly to patient dose and beam quality.

#164 · Safe Work Principles and Radiation Safety

What does collimator accuracy testing confirm?

Answer

That the light field used for positioning matches the actual radiation field within an acceptable tolerance.

#165 · Safe Work Principles and Radiation Safety

What role do technologists play in the QC/QA program, even though they do not perform formal comprehensive equipment evaluations?

Answer

Daily visual and functional checks of equipment, prompt reporting of any malfunction or unexpected image quality change, and cooperation with scheduled formal testing.

#166 · Safe Work Principles and Radiation Safety

Name three sources of ergonomic risk for technologists in medical radiation technology.

Answer

Patient transfers, positioning of heavy or immobile patients, and prolonged standing or repetitive movements such as reaching or lifting portable equipment.

#167 · Safe Work Principles and Radiation Safety

What proper body mechanics reduce the risk of musculoskeletal injury when lifting or transferring a patient?

Answer

Bending at the knees rather than the waist, keeping loads close to the body, using mechanical lift assist devices and transfer aids, and seeking additional staff assistance for difficult transfers.

#168 · Safe Work Principles and Radiation Safety

What does WHMIS stand for, and what does it govern?

Answer

The Workplace Hazardous Materials Information System, which governs the safe handling, storage, and labelling of hazardous chemical products in Canadian workplaces.

#169 • Safe Work Principles and Radiation Safety

What must be readily accessible for every hazardous product used in an imaging department under WHMIS?

Answer

A safety data sheet (SDS), along with standardized labels bearing pictograms that indicate the nature of the hazard.

#170 • Safe Work Principles and Radiation Safety

Why must portable radiographic equipment, lead aprons, and positioning devices be cleaned according to facility protocol between patients?

Answer

Because these items move between patients and can serve as a route of cross-contamination if not properly disinfected, so infection control practices intersect closely with radiation safety in the imaging department.

#171 · Imaging Systems and Equipment

Why is the inside of an X-ray tube kept under vacuum?

Answer

Any residual gas molecules would collide with accelerating electrons, scatter them, and prevent the tube from producing a controlled beam.

#172 · Imaging Systems and Equipment

What material is the X-ray tube filament made of, and why?

Answer

Thoriated tungsten, because tungsten has a high melting point (about 3,422 degrees Celsius) and thoriated tungsten emits electrons more efficiently at a given temperature than pure tungsten.

#173 · Imaging Systems and Equipment

Thermionic emission

Answer

The process by which a heated filament boils off electrons into the space around it, forming a cloud of electrons called the space charge.

#174 · Imaging Systems and Equipment

Scenario: An X-ray tube's cathode has a small filament, a large filament, and a negatively charged, shaped structure surrounding them. What is the function of that structure?

Answer

It keeps the cloud of thermionically emitted electrons condensed into a tight beam as they cross the tube toward the anode.

#175 · Imaging Systems and Equipment

The anode of a general radiography tube serves two purposes. What are they?

Answer

It provides the target material that electrons strike to produce X-rays, and it conducts heat away from that target.

#176 · Imaging Systems and Equipment

Fill in the blank: A rotating anode disc is made of _____ and is typically spun at _____ during an exposure.

Answer

Tungsten-rhenium; 3,000 to 10,000 revolutions per minute.

#177 · Imaging Systems and Equipment

Why does anode rotation allow higher mA and longer exposure times without melting the target?

Answer

Rotation spreads the heat load of each exposure over a much larger physical area of the anode disc than a stationary target could tolerate.

#178 · Imaging Systems and Equipment

Distinguish the actual focal spot from the effective focal spot.

Answer

The actual focal spot is the area of the anode actually struck by the electron beam; the effective focal spot is how that area appears when projected down toward the image receptor.

#179 · Imaging Systems and Equipment

Line focus principle

Answer

The geometric relationship whereby angling the anode makes the effective focal spot appear smaller than the actual focal spot, improving spatial resolution.

#180 · Imaging Systems and Equipment

Scenario: A technologist compares a tube with a steep anode angle to one with a shallow anode angle. What trade-off does each produce?

Answer

A steeper angle gives a smaller effective focal spot and finer resolution but smaller field coverage; a shallower angle gives larger field coverage but is more prone to the anode heel effect.

#181 · Imaging Systems and Equipment

What causes the anode heel effect?

Answer

The angled anode absorbs more of the photons directed toward the anode side of the tube than those directed toward the cathode side, so beam intensity is greater on the cathode side and lesser on the anode side.

#182 · Imaging Systems and Equipment

Scenario: A technologist is imaging a femur, which varies markedly in thickness along the long axis. How should the part be positioned relative to the tube to use the anode heel effect for more uniform density?

Answer

The thicker part should be positioned toward the cathode end of the tube, where beam intensity is higher, and the thinner part toward the anode end, where intensity is lower.

#183 · Imaging Systems and Equipment

List three factors that make the anode heel effect more pronounced.

Answer

A shorter source to image distance, a larger field size, and a smaller anode angle.

#184 · Imaging Systems and Equipment

How are heat units calculated for single-phase X-ray equipment?

Answer

Heat units equal kVp multiplied by mA multiplied by seconds.

#185 · Imaging Systems and Equipment

Why do three-phase and high-frequency generators use a heat unit correction factor > 1?

Answer

These waveforms deliver more usable radiation output per unit of anode heating than single-phase equipment, so a correction factor (1.35 for three-phase six-pulse, 1.41 for three-phase twelve-pulse and high-frequency) accounts for the difference.

#186 · Imaging Systems and Equipment

What are the three major sections of the X-ray circuit?

Answer

The low-voltage (primary) side, the filament circuit, and the high-voltage (secondary) side.

#187 · Imaging Systems and Equipment

What is the role of the autotransformer in the X-ray circuit?

Answer

It allows the operator's console kVp selection to adjust the voltage sent forward to the step-up transformer.

#188 · Imaging Systems and Equipment

Fill in the blank: The step-up transformer increases voltage by a fixed ratio, commonly _____, to produce diagnostic-range kilovoltage.

Answer

500:1

#189 · Imaging Systems and Equipment

Why must the high voltage from the step-up transformer be rectified?

Answer

Standard electrical supply is alternating current, so it must be converted into a waveform that flows in one direction only, cathode to anode, for the tube to function properly.

#190 · Imaging Systems and Equipment

How does full-wave rectification differ from older half-wave rectification?

Answer

Full-wave rectification uses four diodes so both the positive and negative halves of the AC cycle produce useful X-ray output, rather than wasting half the cycle as half-wave rectified units did.

#191 · Imaging Systems and Equipment

Ripple

Answer

The degree to which a generator's voltage waveform drops from peak toward zero during a cycle; lower ripple means a waveform closer to constant potential.

#192 · Imaging Systems and Equipment

Scenario: A department upgrades from single-phase to high-frequency generator equipment. What change in image production should the technologist expect at the same nominal exposure factors?

Answer

Higher average photon energy and greater X-ray output per mAs, with much shorter exposure times possible, because the nearly constant potential waveform keeps the tube operating near peak kVp throughout the exposure.

#193 · Imaging Systems and Equipment

Fill in the blank: Twelve-pulse three-phase generators have ripple under _____ percent, while high-frequency generators have ripple often below _____ percent.

Answer

4 percent; 1 percent.

#194 · Imaging Systems and Equipment

What is the primary controller of the quantity of X-rays produced, and how does it affect receptor exposure?

Answer

Milliamperere-seconds (mAs); doubling mAs doubles the number of photons and directly doubles receptor exposure, all other factors held constant.

#195 · Imaging Systems and Equipment

Scenario: A technologist uses 200 mA at 0.05 seconds for one exposure and 100 mA at 0.1 seconds for another. What is true about these two exposures?

Answer

Both yield 10 mAs, so they produce the same overall exposure, though the shorter-time combination helps reduce motion artifact.

#196 · Imaging Systems and Equipment

The 15 percent rule

Answer

Increasing kVp by 15 percent produces approximately the same increase in receptor exposure as doubling the mAs, and decreasing kVp by 15 percent halves receptor exposure similarly to halving the mAs.

#197 · Imaging Systems and Equipment

Why does raising kVp while proportionally lowering mAs (per the 15 percent rule) reduce patient dose but also lower subject contrast?

Answer

It delivers less overall dose for an equivalent receptor exposure, but higher-energy beams produce more Compton scatter relative to photoelectric absorption, giving less differential attenuation between tissues.

#198 · Imaging Systems and Equipment

Inverse square law

Answer

The intensity of radiation is inversely proportional to the square of the distance from the source; doubling source to image distance reduces exposure to one quarter, and halving it increases exposure to four times.

#199 · Imaging Systems and Equipment

What is the purpose of a backup manual timer alongside automatic exposure control?

Answer

It protects the tube and patient from excessive exposure if the AEC circuit fails to terminate the exposure appropriately.

#200 · Imaging Systems and Equipment

Describe how a computed radiography (CR) plate captures a latent image.

Answer

X-rays raise electrons within a photostimulable phosphor plate to a higher, metastable energy state where they remain trapped, forming a latent image until the plate is read.

#201 · Imaging Systems and Equipment

Photostimulable luminescence

Answer

The release of stored energy as visible light when a laser scans a CR phosphor plate, stimulating trapped electrons; a photomultiplier tube detects this light and converts it into an electronic signal.

#202 · Imaging Systems and Equipment

Why must a CR plate be erased with bright light before reuse?

Answer

To clear any residual trapped signal left in the phosphor before the plate is exposed again.

#203 · Imaging Systems and Equipment

Distinguish indirect conversion and direct conversion DR detectors.

Answer

Indirect conversion detectors use a scintillator (cesium iodide or gadolinium oxysulfide) to convert X-rays to light, detected by a photodiode array; direct conversion detectors use a photoconductor (amorphous selenium) to convert X-rays directly to an electronic charge with no light step.

#204 · Imaging Systems and Equipment

Why do direct conversion DR detectors generally provide slightly sharper spatial resolution than indirect conversion detectors?

Answer

There is no intermediate light generation, so there is no lateral light spread within the detector layer.

#205 · Imaging Systems and Equipment

Scenario: A digital radiograph looks acceptable on the display, but the technologist wants to confirm the correct amount of radiation reached the detector. What should they check, and why?

Answer

The exposure index and deviation index, because post-processing automatically adjusts displayed brightness and contrast regardless of actual exposure, so visual inspection alone is unreliable.

#206 · Imaging Systems and Equipment

Deviation index (DI)

Answer

A logarithmic value expressing how far the actual exposure index deviates from the target exposure index for an examination; zero means exposure matched the target, positive values indicate overexposure, negative values indicate underexposure.

#207 · Imaging Systems and Equipment

Fill in the blank: A deviation index outside $\pm$ _____ typically indicates a significant technical error requiring investigation and often a repeat exposure.

Answer

3

#208 · Imaging Systems and Equipment

Why are exposure index values not directly comparable across different manufacturers' equipment?

Answer

Exposure index calculation methods differ between manufacturers, and even between different systems from the same manufacturer, so technologists must learn the target EI range specific to their own department's equipment.

#209 · Imaging Systems and Equipment

What risk does inadequate exposure indicator checking create when transitioning from film to digital imaging?

Answer

Unrecognized dose creep, because automatic brightness and contrast processing can visually mask a significantly overexposed image, hiding excessive patient dose from casual review.

#210 · Imaging Systems and Equipment

How does a grid improve image contrast?

Answer

It absorbs scattered radiation before it reaches the image receptor, reducing the fogging effect that scatter produces on the finished image.

#211 · Clinical Principles Procedures and Contrast Pharmaceuticals

What are the two major divisions of the skeleton?

Answer

The axial skeleton (skull, vertebral column, ribs, and sternum) and the appendicular skeleton (the limbs plus the pectoral and pelvic girdles).

#212 · Clinical Principles Procedures and Contrast Pharmaceuticals

How many bones make up the cranium, and what are they?

Answer

Eight bones: the single frontal, occipital, sphenoid, and ethmoid bones, plus the paired parietal and temporal bones.

#213 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why is C7 called the vertebra prominens?

Answer

Its long spinous process is easily palpable at the base of the neck, making it a reliable surface landmark for locating the cervicothoracic junction.

#214 · Clinical Principles Procedures and Contrast Pharmaceuticals

What distinguishes the atlas (C1) from a typical vertebra?

Answer

The atlas has no vertebral body and articulates with the occipital condyles, permitting nodding of the head.

#215 · Clinical Principles Procedures and Contrast Pharmaceuticals

What structure on the axis (C2) allows rotation of the head?

Answer

The dens, or odontoid process, a superior projection around which the atlas rotates.

#216 · Clinical Principles Procedures and Contrast Pharmaceuticals

At what vertebral level does the jugular (suprasternal) notch lie?

Answer

Approximately T2 to T3, marking the superior border of the manubrium.

#217 · Clinical Principles Procedures and Contrast Pharmaceuticals

At what vertebral level does the xiphoid process lie, and what does it approximate?

Answer

Approximately T9 to T10, roughly corresponding to the inferior border of the heart.

#218 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is the sternal angle (angle of Louis) and why is it useful?

Answer

The palpable ridge where the manubrium meets the sternal body at T4 to T5, marking the attachment of the second rib and serving as a rib-counting landmark.

#219 · Clinical Principles Procedures and Contrast Pharmaceuticals

At what vertebral level does the iliac crest lie?

Answer

Approximately L4 to L5, making it a critical landmark for centring lumbar spine and abdominal projections.

#220 · Clinical Principles Procedures and Contrast Pharmaceuticals

What landmark is used to centre hip radiographs?

Answer

The greater trochanter of the femur, palpable at the level of the symphysis pubis.

#221 · Clinical Principles Procedures and Contrast Pharmaceuticals

List the five bone shape classifications with an example of each.

Answer

Long (femur), short (carpals), flat (sternum), irregular (vertebrae), and sesamoid (patella).

#222 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is the difference between intramembranous and endochondral ossification?

Answer

Intramembranous ossification forms flat bones directly from mesenchymal tissue with no cartilage stage, while endochondral ossification forms most other bones from a cartilage model progressively replaced by bone.

#223 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why must a paediatric growth plate never be mistaken for a fracture?

Answer

The epiphyseal plate is a normal radiolucent band of hyaline cartilage where longitudinal growth occurs until it ossifies into the epiphyseal line, typically in the late teens to early twenties.

#224 · Clinical Principles Procedures and Contrast Pharmaceuticals

At what level does the trachea bifurcate into the main bronchi, and what is that point called?

Answer

Approximately T4 to T5, at a point called the carina.

#225 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why do aspirated foreign bodies more commonly lodge in the right lung?

Answer

The right main bronchus is shorter, wider, and more vertical, branching at roughly 25 degrees from vertical compared to the left bronchus's 45 degrees.

#226 · Clinical Principles Procedures and Contrast Pharmaceuticals

How many lobes does the right lung have compared to the left, and why the difference?

Answer

The right lung has three lobes separated by horizontal and oblique fissures; the left has only two lobes since the cardiac notch occupies the space a middle lobe would otherwise fill.

#227 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is the difference between visceral and parietal pleura?

Answer

The visceral pleura adheres directly to the lung surface, while the parietal pleura lines the thoracic wall, diaphragm, and mediastinum, with the pleural cavity between them.

#228 · Clinical Principles Procedures and Contrast Pharmaceuticals

What structures form the right and left borders of the heart on a PA chest radiograph?

Answer

The right border is formed by the right atrium; the left border is formed, superior to inferior, by the aortic knob, main pulmonary artery segment, left atrial appendage, and left ventricle.

#229 · Clinical Principles Procedures and Contrast Pharmaceuticals

What three arteries arise from the aortic arch?

Answer

The brachiocephalic trunk, the left common carotid artery, and the left subclavian artery.

#230 · Clinical Principles Procedures and Contrast Pharmaceuticals

What are the four regions of the stomach?

Answer

The cardia, fundus, body, and pylorus.

#231 · Clinical Principles Procedures and Contrast Pharmaceuticals

What are the three parts of the small intestine, in order?

Answer

The duodenum, jejunum, and ileum.

#232 · Clinical Principles Procedures and Contrast Pharmaceuticals

What anatomic feature radiographically distinguishes the large intestine from the small intestine?

Answer

Haustra, the sacculated pouches created by the taeniae coli, and its generally more peripheral location framing the small bowel.

#233 · Clinical Principles Procedures and Contrast Pharmaceuticals

Describe the dual blood supply of the liver.

Answer

The hepatic artery delivers oxygenated blood from the celiac trunk, while the portal vein delivers nutrient-rich but deoxygenated blood from the gastrointestinal tract and spleen.

#234 · Clinical Principles Procedures and Contrast Pharmaceuticals

What vertebral levels typically span the kidneys, and why does the right kidney sit slightly lower?

Answer

Approximately T12 to L3; the right kidney sits lower because of the space occupied superiorly by the liver.

#235 · Clinical Principles Procedures and Contrast Pharmaceuticals

On a standard axial CT image, which side of the patient appears on the viewer's left?

Answer

The patient's right side, since axial CT images are displayed as though viewed from the patient's feet looking toward the head, matching the AP radiograph convention.

#236 · Clinical Principles Procedures and Contrast Pharmaceuticals

What SID is conventionally used for chest radiography, and why is it longer than the standard 100 cm used elsewhere?

Answer

180 cm, used specifically to minimize magnification of the cardiac silhouette, since the heart lies close to the anterior chest wall.

#237 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why is a PA chest projection preferred over an AP for assessing true cardiac size?

Answer

In a PA projection the heart lies closer to the image receptor, minimizing the magnification that occurs with increased part-to-image-distance.

#238 · Clinical Principles Procedures and Contrast Pharmaceuticals

Where is the central ray directed for a standard PA chest projection?

Answer

Horizontally and perpendicular to the receptor, entering at the level of T7, approximately the inferior angle of the scapula.

#239 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why is full inspiration essential for a diagnostic PA chest radiograph?

Answer

It maximizes lung aeration, lowers the diaphragm to roughly the tenth posterior rib, and provides the greatest contrast between aerated lung and any underlying pathology; poor inspiration falsely enlarges the cardiac silhouette.

#240 · Clinical Principles Procedures and Contrast Pharmaceuticals

How many posterior ribs should be visible above the diaphragm on a well-inspired PA chest film?

Answer

Ten posterior ribs.

#241 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why does chest radiography use a high kVp technique?

Answer

High kVp (commonly 110 to 125) produces a long grey scale that visualizes both dense mediastinal structures and low-density aerated lung on one exposure, and the correspondingly low mAs minimizes motion blur.

#242 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why is the lateral chest projection conventionally performed left side against the receptor?

Answer

Because the heart lies predominantly in the left hemithorax, imaging the left side first minimizes cardiac magnification.

#243 · Clinical Principles Procedures and Contrast Pharmaceuticals

What evaluation criterion confirms minimal rotation on a lateral chest radiograph?

Answer

The posterior ribs should be superimposed within approximately one centimetre of each other.

#244 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why does a portable AP chest radiograph magnify the cardiac silhouette more than a departmental PA study?

Answer

The anterior chest and heart lie further from the receptor in an AP projection, and SID is often reduced on portable equipment, both increasing magnification.

#245 · Clinical Principles Procedures and Contrast Pharmaceuticals

How is rotation assessed on a portable AP chest radiograph?

Answer

By comparing the distances from the spinous processes to each medial clavicular end, which should be equal in a non-rotated study.

#246 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is an air bronchogram and what does it indicate?

Answer

Branching lucent air-filled bronchi seen against surrounding white consolidated lung tissue, confirming the opacity originates within the lung parenchyma rather than the pleural space.

#247 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is the silhouette sign?

Answer

Loss of a normally sharp anatomical border, such as the heart border or a hemidiaphragm, indicating the adjacent lung tissue has lost its air content, which helps localize the affected lobe.

#248 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is the classic radiographic appearance of pneumothorax?

Answer

A thin, sharp visceral pleural line running parallel to the chest wall with a complete absence of lung markings peripheral to that line.

#249 · Clinical Principles Procedures and Contrast Pharmaceuticals

What signs indicate a tension pneumothorax?

Answer

Mediastinal shift away from the affected side, flattening or inversion of the ipsilateral hemidiaphragm, and rib spreading on the affected side; this is a life-threatening emergency requiring immediate physician notification.

#250 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is the meniscus sign?

Answer

A curved, upward-rising fluid opacity along the lateral chest wall, seen with pleural effusion as fluid layers in the dependent thorax and appears higher laterally than medially.

#251 · Clinical Principles Procedures and Contrast Pharmaceuticals

Which view is most sensitive for detecting a small pleural effusion?

Answer

A lateral decubitus view with the affected side down, since free-flowing fluid layers along the dependent lateral chest wall.

#252 · Clinical Principles Procedures and Contrast Pharmaceuticals

How is atelectasis distinguished radiographically from pneumonia?

Answer

Atelectasis shows volume loss signs such as fissure displacement, hemidiaphragm elevation, mediastinal shift toward the affected side, and rib crowding, while pneumonia typically shows normal or slightly increased lung volume.

#253 · Clinical Principles Procedures and Contrast Pharmaceuticals

What are Kerley B lines and what do they indicate?

Answer

Short, horizontal linear opacities at the lung bases representing fluid within interlobular septa, an early sign of interstitial pulmonary edema.

#254 · Clinical Principles Procedures and Contrast Pharmaceuticals

Describe the radiographic pattern of alveolar pulmonary edema.

Answer

Bilateral, symmetric, perihilar opacities in a "bat wing" or "butterfly" distribution, radiating outward from the hila while relatively sparing the lung periphery.

#255 · Clinical Principles Procedures and Contrast Pharmaceuticals

How is the cardiothoracic ratio calculated, and what value is generally abnormal?

Answer

Maximum transverse cardiac diameter divided by maximum internal thoracic diameter at the diaphragm level; a ratio exceeding 50 percent on a true PA film at full inspiration is generally abnormal.

#256 · Clinical Principles Procedures and Contrast Pharmaceuticals

Where is the central ray centred for a supine KUB radiograph?

Answer

Perpendicular to the receptor, entering at the level of the iliac crest, which approximates L4.

#257 · Clinical Principles Procedures and Contrast Pharmaceuticals

What evaluation criterion confirms an unrotated supine abdomen radiograph?

Answer

Symmetrical iliac wings and obturator foramina.

#258 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why is an upright or decubitus abdomen view essential in addition to a supine view when perforation is suspected?

Answer

A supine radiograph alone can miss even substantial amounts of free air; free air rises and is only reliably detected beneath the diaphragm on an upright view or between the liver and abdominal wall on a decubitus view.

#259 · Clinical Principles Procedures and Contrast Pharmaceuticals

Where does free intraperitoneal air characteristically appear on an upright chest or abdomen radiograph?

Answer

As a thin curvilinear lucency beneath the right hemidiaphragm, since the gastric bubble can mimic free air on the left.

#260 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why is the left lateral decubitus position preferred over the right when evaluating for free air?

Answer

With the patient on the left side, free air rises and collects along the right side adjacent to the liver, where it is more easily distinguished from the gastric bubble.

#261 • Clinical Principles Procedures and Contrast Pharmaceuticals

How long should a patient lie in the left lateral decubitus position before an abdomen exposure for suspected free air?

Answer

A minimum of five to ten minutes, to allow free air time to rise and collect along the right side.

#262 • Clinical Principles Procedures and Contrast Pharmaceuticals

What three views typically make up an acute abdomen (obstruction) series?

Answer

A supine abdomen, an upright or decubitus abdomen, and often a PA chest radiograph.

#263 · Clinical Principles Procedures and Contrast Pharmaceuticals

What radiographic feature distinguishes small bowel from large bowel obstruction?

Answer

Small bowel obstruction shows centrally distributed dilated loops with valvulae conniventes that fully traverse the lumen, while large bowel obstruction shows peripherally distributed dilated loops with haustra that do not fully traverse the lumen.

#264 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is the "stepladder" sign?

Answer

Multiple air-fluid levels at differing heights within dilated small bowel loops on an upright abdomen film, a classic sign of small bowel obstruction.

#265 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is a phlebolith and why is it clinically insignificant?

Answer

A small, rounded pelvic calcification with a lucent centre representing calcified thrombus within a pelvic vein, generally an incidental finding of no clinical significance.

#266 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why are most gallstones not visible on a plain abdominal radiograph?

Answer

Most gallstones have insufficient calcium content to be radiopaque, so only a minority are visible on plain film.

#267 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why must some abdominal foreign bodies (like most plastics) be evaluated with caution on plain film?

Answer

Radiolucent foreign bodies are not visible on plain radiography, so a negative plain film does not rule out ingestion or retention of a non-radiopaque object.

#268 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is the purpose of the ALARA principle when deciding whether to repeat a non-diagnostic chest or abdomen film?

Answer

It balances keeping patient dose as low as reasonably achievable against the clinical need for a diagnostic image, since a non-diagnostic study that must be repeated adds dose with no diagnostic benefit from the original exposure.

#269 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why is confirming or reasonably excluding pregnancy important before an abdominal or pelvic radiographic examination?

Answer

It is a standard radiation protection step in female patients of childbearing potential, following facility pregnancy-screening policy, before proceeding with an abdominal or pelvic exposure.

#270 · Clinical Principles Procedures and Contrast Pharmaceuticals

Where is the central ray centred for an upright abdomen radiograph, and why?

Answer

Approximately 5 cm above the iliac crest, to ensure the diaphragm is included since free air rises and collects beneath it in an upright patient.

#271 · Clinical Principles Procedures and Contrast Pharmaceuticals

What three standard projections make up a routine hand series?

Answer

PA, PA oblique, and lateral.

#272 · Clinical Principles Procedures and Contrast Pharmaceuticals

What does the PA oblique hand projection demonstrate that the true PA cannot?

Answer

It opens the interphalangeal joint spaces that overlap on a true PA and demonstrates the phalanges with less superimposition.

#273 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is the ulnar deviation PA wrist ("scaphoid view") used for?

Answer

It elongates the scaphoid and reduces foreshortening when a scaphoid fracture is clinically suspected.

#274 · Clinical Principles Procedures and Contrast Pharmaceuticals

What landmark confirms a true lateral wrist with no rotation?

Answer

Perfect superimposition of the radius and ulna, with the pisiform sitting roughly in the middle third between the distal radius and the flexor surface of the scaphoid.

#275 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why must both the wrist and elbow joints be included on a forearm radiograph?

Answer

To rule out an associated Monteggia fracture-dislocation (ulna fracture with radial head dislocation) or Galeazzi fracture-dislocation (radius fracture with distal radioulnar joint dislocation).

#276 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is the "sail sign" on a lateral elbow radiograph?

Answer

An elevated, visible anterior fat pad, a thin radiolucent triangle pressed away from the humerus by joint effusion, most classically from an occult radial head fracture.

#277 · Clinical Principles Procedures and Contrast Pharmaceuticals

Where is the posterior fat pad located in a healthy elbow, and what does its visibility suggest?

Answer

In a healthy elbow the posterior fat pad is hidden within the olecranon fossa and not visible; any visible posterior fat pad suggests a joint effusion, often from occult fracture.

#278 · Clinical Principles Procedures and Contrast Pharmaceuticals

How should rotation of the humerus be achieved between AP and lateral projections?

Answer

By internally or externally rotating the entire arm from the shoulder, never by rotating only the forearm, since the humerus must move as a single unit for the projections to be truly orthogonal.

#279 · Clinical Principles Procedures and Contrast Pharmaceuticals

What tubercle is seen in profile on an externally rotated AP shoulder, and what about internal rotation?

Answer

External rotation places the greater tubercle in profile laterally; internal rotation places the lesser tubercle in profile medially.

#280 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why is the scapular Y projection valuable in acute shoulder trauma?

Answer

It reliably identifies anterior or posterior glenohumeral dislocation, since the humeral head normally superimposes on the junction of the Y, without requiring the arm to be rotated.

#281 · Clinical Principles Procedures and Contrast Pharmaceuticals

How much is the central ray angled for an AP foot projection, and why?

Answer

10 to 15 degrees posteriorly (toward the heel), to compensate for the longitudinal arch and open the tarsometatarsal joint spaces.

#282 · Clinical Principles Procedures and Contrast Pharmaceuticals

What does the medial oblique foot projection best demonstrate?

Answer

The sinus tarsi and the fifth metatarsal tuberosity, an area of frequent avulsion fracture in inversion ankle injuries.

#283 · Clinical Principles Procedures and Contrast Pharmaceuticals

How is the mortise ankle projection obtained, and what does it demonstrate?

Answer

By internally rotating the leg 15 to 20 degrees, placing the malleoli parallel to the receptor and opening the ankle mortise uniformly on all three sides for joint space assessment.

#284 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why must both the knee and ankle joints be included on a tibia-fibula radiograph?

Answer

As with the forearm, both adjacent joints must be demonstrated to rule out associated injury at either end of the paired long bones.

#285 · Clinical Principles Procedures and Contrast Pharmaceuticals

Where is the central ray centred for an AP knee projection?

Answer

Half an inch below the apex of the patella.

#286 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is a lipohemarthrosis and how is it best demonstrated?

Answer

A fat-fluid level within the knee joint capsule strongly suggesting an intra-articular fracture; it is best seen on a horizontal beam lateral knee projection.

#287 · Clinical Principles Procedures and Contrast Pharmaceuticals

What are the tunnel (Camp Coventry) and sunrise (Merchant) knee projections used to evaluate?

Answer

The PA axial tunnel projection evaluates the intercondylar fossa; the axial sunrise projection evaluates the patellofemoral joint.

#288 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why is a cross-table (Danelius-Miller) lateral mandatory, not a frog-leg lateral, when a hip fracture is suspected?

Answer

The frog-leg position requires external rotation and abduction, which can displace an impacted femoral neck fracture and convert a stable injury into an unstable one.

#289 · Clinical Principles Procedures and Contrast Pharmaceuticals

Where is the central ray centred for an AP pelvis projection?

Answer

Midway between the anterior superior iliac spines at the level of the greater trochanters, roughly 5 cm superior to the pubic symphysis.

#290 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why are the legs internally rotated 15 to 20 degrees for an AP pelvis or AP femur projection?

Answer

To place the femoral necks parallel to the image receptor, reducing foreshortening.

#291 · Clinical Principles Procedures and Contrast Pharmaceuticals

Define a comminuted fracture.

Answer

A fracture consisting of more than two fragments, often produced by high-energy trauma.

#292 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is a greenstick fracture, and in whom is it seen?

Answer

An incomplete fracture where the bone bends and cracks on the convex side while the concave cortex stays intact; it occurs almost exclusively in the more pliable bone of children.

#293 · Clinical Principles Procedures and Contrast Pharmaceuticals

Compare transverse, oblique, and spiral fracture lines.

Answer

A transverse fracture runs perpendicular to the long axis, an oblique fracture runs at an angle across the long axis, and a spiral fracture wraps helically around the shaft from a rotational force.

#294 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is an avulsion fracture, and where is it commonly seen?

Answer

A small bone fragment pulled away at a tendon or ligament attachment site, commonly at the base of the fifth metatarsal or the anterior tibial spine.

#295 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is a pathologic fracture?

Answer

A fracture occurring through bone already weakened by an underlying disease process, such as a metastatic lesion, bone tumour, or severe osteoporosis, typically from a mechanism that would not fracture normal bone.

#296 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is a stress fracture, and why can it be radiographically invisible early on?

Answer

A fracture developing gradually from repetitive submaximal loading rather than a single trauma; it may be subtle or entirely invisible for the first one to two weeks, sometimes requiring bone scan or MRI for early detection.

#297 · Clinical Principles Procedures and Contrast Pharmaceuticals

Describe a Salter-Harris Type I fracture.

Answer

A fracture through the physis alone with the epiphysis cleanly separating from the metaphysis; plain radiographs can appear entirely normal if the fragments are not displaced.

#298 · Clinical Principles Procedures and Contrast Pharmaceuticals

Describe a Salter-Harris Type II fracture.

Answer

The most common pattern, extending through the physis and exiting through a triangular fragment of metaphysis.

#299 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why does a Salter-Harris Type V fracture carry the highest risk of growth arrest?

Answer

It is a crush injury to the physis itself with no obvious displaced fracture line, frequently missed on initial imaging, and directly compresses the germinal layer of the growth plate.

#300 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is the rule of right angles (rule of two projections)?

Answer

A minimum of two projections obtained at 90 degrees to each other are required to fully characterize a fracture or dislocation in three dimensions.

#301 · Clinical Principles Procedures and Contrast Pharmaceuticals

Explain the trauma "rule of twos."

Answer

Two views at right angles, two joints included for long bone shafts, two limbs compared when a finding is uncertain, and two occasions, meaning a follow-up film in 10 to 14 days if initial films are equivocal.

#302 · Clinical Principles Procedures and Contrast Pharmaceuticals

How should a technologist adapt positioning for a trauma patient who cannot be moved into a standard position?

Answer

By modifying tube angle, receptor angle, or patient support with sponges or angle blocks, never by manipulating a suspected fracture to achieve ideal positioning.

#303 · Clinical Principles Procedures and Contrast Pharmaceuticals

What must a lateral cervical spine radiograph demonstrate, and why?

Answer

All seven cervical vertebrae plus the C7-T1 junction, since a fracture at the cervicothoracic junction is easily missed if the lower cervical spine is cut off.

#304 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is a swimmer's projection used for?

Answer

It is used in cervical spine imaging, with one arm raised overhead to depress the shoulder and visualize the cervicothoracic junction when it is obscured on the standard lateral.

#305 · Clinical Principles Procedures and Contrast Pharmaceuticals

What does the AP open mouth (odontoid) projection demonstrate, and what fractures is it used to evaluate?

Answer

It demonstrates the dens and the lateral masses of C1 in relation to C2, used to evaluate Jefferson fracture of the atlas or odontoid fracture.

#306 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why is a breathing technique sometimes used for the lateral thoracic spine?

Answer

A long exposure during shallow continuous respiration blurs overlying rib and lung markings while keeping the vertebral bodies relatively sharp, since the spine moves comparatively little during quiet breathing.

#307 · Clinical Principles Procedures and Contrast Pharmaceuticals

What does the "Scotty dog" appearance on a lumbar oblique radiograph help detect?

Answer

Spondylolysis, a defect of the pars interarticularis.

#308 · Clinical Principles Procedures and Contrast Pharmaceuticals

How does a compression fracture appear on a lateral spine radiograph?

Answer

As a wedge-shaped vertebra with reduced anterior vertebral body height relative to the posterior height, commonly from osteoporotic collapse or axial loading trauma.

#309 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is spondylolisthesis?

Answer

Anterior slippage of one vertebral body relative to the one below it, most often at L4-L5 or L5-S1, graded by the percentage of anterior displacement and best seen on the lateral projection.

#310 · Clinical Principles Procedures and Contrast Pharmaceuticals

How does osteoarthritis differ radiographically from rheumatoid arthritis?

Answer

Osteoarthritis shows asymmetric joint space narrowing, subchondral sclerosis, and osteophyte formation, while rheumatoid arthritis shows symmetric joint space narrowing, periarticular osteopenia, and marginal erosions.

#311 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is the difference between positive and negative contrast agents?

Answer

Positive agents have a higher atomic number than soft tissue and appear white (radiopaque); negative agents have a lower atomic number and appear black (radiolucent).

#312 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why does iodine make an effective positive contrast agent?

Answer

Iodine has an atomic number of 53, giving it a k-edge close to the diagnostic imaging energy range, making it an efficient x-ray attenuator.

#313 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why do non-ionic iodinated contrast agents cause fewer adverse reactions than ionic agents?

Answer

Non-ionic agents do not dissociate into charged particles in solution, producing lower osmolality per unit of iodine and therefore fewer physiologic effects and reactions.

#314 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why is barium sulfate dangerous if it escapes into the peritoneal cavity?

Answer

It can incite a severe inflammatory reaction called barium peritonitis, since it is not readily absorbed or cleared once outside the intact gut.

#315 • Clinical Principles Procedures and Contrast Pharmaceuticals

Why is carbon dioxide sometimes preferred over room air as a negative contrast agent?

Answer

Carbon dioxide is highly soluble in blood and is rapidly absorbed and exhaled, reducing the risk of air embolism in vascular applications.

#316 • Clinical Principles Procedures and Contrast Pharmaceuticals

How is iodinated contrast media eliminated from the body in a patient with normal renal function?

Answer

Almost entirely through renal filtration by the glomerulus, with most of the dose cleared from plasma and excreted in urine within a few hours.

#317 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is the single strongest predictor of a future contrast reaction?

Answer

A documented prior reaction to iodinated contrast media.

#318 · Clinical Principles Procedures and Contrast Pharmaceuticals

Does a shellfish allergy indicate increased risk of an iodinated contrast reaction?

Answer

No. The historical belief linking shellfish allergy to iodine allergy has been discredited; a shellfish allergy confers no greater risk than any other food allergy.

#319 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why are β -blocking medications a notable risk factor during a contrast reaction?

Answer

They can blunt the physiologic response to epinephrine, making a reaction harder to treat, and can mask the tachycardic warning sign that often precedes hypotension.

#320 · Clinical Principles Procedures and Contrast Pharmaceuticals

Describe a typical premedication protocol for a patient with a documented prior contrast reaction.

Answer

An oral corticosteroid such as prednisone given at scheduled intervals beginning the day before the study, combined with an antihistamine such as diphenhydramine shortly before injection; this reduces but does not eliminate the risk of a repeat reaction.

#321 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is contrast-induced nephropathy, and how is it defined?

Answer

An acute decline in renal function after intravascular iodinated contrast, generally defined by a measurable rise in serum creatinine within 48 to 72 hours of exposure.

#322 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why is eGFR preferred over creatinine alone for renal function screening before contrast?

Answer

Creatinine alone is affected by muscle mass, age, and sex; eGFR incorporates these factors along with creatinine to give a more reliable estimate of kidney function across different patients.

#323 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why does metformin require special consideration before iodinated contrast administration?

Answer

In a patient with significantly impaired renal function, metformin can accumulate to dangerous levels if renal function deteriorates further after contrast, raising the risk of lactic acidosis.

#324 · Clinical Principles Procedures and Contrast Pharmaceuticals

When is a water-soluble contrast agent used instead of barium for a gastrointestinal study?

Answer

When bowel perforation is clinically suspected, since water-soluble agents are safely absorbed and excreted if they leak into the peritoneal cavity, unlike barium.

#325 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why is the median cubital vein generally the preferred first-choice site for venipuncture?

Answer

It tends to be large, well anchored by surrounding connective tissue, and relatively superficial, making it comfortable for the patient and reliable for cannulation.

#326 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why is the basilic vein a less favoured first choice for venipuncture despite often being the largest antecubital vein?

Answer

It lies close to the brachial artery and median nerve and tends to roll more readily under the needle, increasing the risk of arterial puncture or nerve injury.

#327 · Clinical Principles Procedures and Contrast Pharmaceuticals

What catheter gauge is commonly used for power-injected CT contrast, and why?

Answer

An 18 to 20 gauge catheter, selected to withstand the higher flow rates and pressures generated by a power injector.

#328 · Clinical Principles Procedures and Contrast Pharmaceuticals

At what angle is the needle inserted relative to the skin during standard venipuncture technique?

Answer

Approximately 15 to 30 degrees, bevel up, following the direction of the vein.

#329 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is the difference between infiltration and extravasation?

Answer

Infiltration is the leakage of a non-vesicant fluid such as plain saline into subcutaneous tissue, while extravasation refers specifically to leakage of contrast media or another vesicant or irritant substance.

#330 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is often the earliest and most sensitive sign of contrast extravasation during a power injection?

Answer

Patient-reported pain at the injection site, which frequently precedes any visible swelling or injector pressure alarm.

#331 · Clinical Principles Procedures and Contrast Pharmaceuticals

How is a hematoma from venipuncture best prevented?

Answer

Applying firm, direct pressure over the puncture site for an adequate period after needle or catheter removal.

#332 · Clinical Principles Procedures and Contrast Pharmaceuticals

How is inadvertent arterial puncture recognized during venipuncture?

Answer

By bright red, pulsatile blood return under higher pressure than typical venous flow, requiring immediate needle withdrawal and sustained direct pressure.

#333 · Clinical Principles Procedures and Contrast Pharmaceuticals

Contrast a mild versus a moderate contrast reaction.

Answer

A mild reaction includes limited, self-resolving signs like a few hives or transient nausea with no airway or circulatory compromise; a moderate reaction includes more pronounced signs such as widespread urticaria, mild bronchospasm, or transient hypotension and requires active treatment.

#334 · Clinical Principles Procedures and Contrast Pharmaceuticals

What defines a severe contrast reaction, and what is the required immediate response?

Answer

Life-threatening signs such as airway compromise, hypotensive shock, or cardiac arrest; the response is immediate activation of the emergency response system, epinephrine per protocol, high-flow oxygen, and basic life support as needed.

#335 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why should the intravenous line be kept in place and patent during a contrast reaction?

Answer

To allow rapid administration of emergency medications if the reaction escalates.

#336 · Clinical Principles Procedures and Contrast Pharmaceuticals

What are the two fixed reference points on the Hounsfield unit scale?

Answer

Water at 0 Hounsfield units and air at negative 1000 Hounsfield units.

#337 · Clinical Principles Procedures and Contrast Pharmaceuticals

How do window width and window level differ in CT image display?

Answer

Window width determines the total range of Hounsfield units displayed as greyscale; window level (window centre) determines the midpoint of that range.

#338 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why can a CT dataset be rewinded repeatedly without additional radiation exposure?

Answer

The underlying attenuation data does not change; only the way it is displayed changes.

#339 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is pitch in helical CT, and what does a pitch of 1 mean?

Answer

Pitch is table travel per gantry rotation divided by total beam width; a pitch of 1 means the table moves exactly the beam width per rotation, giving contiguous coverage with no gaps or overlap.

#340 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is the trade-off of using a pitch > 1 in helical CT?

Answer

It speeds up the scan and reduces patient dose but leaves small gaps in raw data that must be interpolated.

#341 · Clinical Principles Procedures and Contrast Pharmaceuticals

Why is non-contrast head CT the first-line study for suspected acute stroke or intracranial haemorrhage?

Answer

Fresh blood is hyperdense relative to normal brain tissue and is readily distinguished from brain parenchyma on an unenhanced scan.

#342 · Clinical Principles Procedures and Contrast Pharmaceuticals

How does bolus tracking work for CT pulmonary angiography?

Answer

Software monitors contrast arrival in the main pulmonary artery and automatically triggers the scan once a threshold density is reached, typically 15 to 25 seconds after injection begins.

#343 · Clinical Principles Procedures and Contrast Pharmaceuticals

In a multiphase liver CT protocol, what does the arterial phase capture and when is it acquired?

Answer

It captures hypervascular lesions such as hepatocellular carcinoma at their brightest, acquired around 25 to 35 seconds after injection.

#344 · Clinical Principles Procedures and Contrast Pharmaceuticals

What does CT urography's delayed excretory phase capture, and when is it acquired?

Answer

It captures contrast excreted into the renal collecting systems, ureters, and bladder, typically acquired 5 to 15 minutes after injection.

#345 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is automatic tube current modulation in CT, and why is it used?

Answer

It adjusts the milliamperage in real time based on patient thickness and attenuation as the tube rotates and moves along the patient, reducing dose while maintaining image quality.

#346 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is pulsed fluoroscopy, and why does it reduce patient dose?

Answer

The x-ray beam is delivered in short synchronized bursts rather than continuously, so total dose is substantially reduced compared to continuous fluoroscopy, especially at lower pulse rates.

#347 · Clinical Principles Procedures and Contrast Pharmaceuticals

How do flat panel detectors differ from image intensifiers in fluoroscopy?

Answer

Flat panel detectors produce a digital image without geometric distortion, in a thinner, lighter housing, with improved dynamic range, unlike image intensifiers, which suffer from pincushion distortion at the periphery.

#348 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is myelography, and what technique is used to administer the contrast?

Answer

The injection of water-soluble iodinated contrast into the subarachnoid space via lumbar puncture, followed by fluoroscopic observation to outline the spinal cord, nerve roots, and thecal sac.

#349 · Clinical Principles Procedures and Contrast Pharmaceuticals

What vascular access technique is used for diagnostic angiography?

Answer

The Seldinger technique: needle puncture, guidewire insertion, and catheter exchange over the wire, most commonly via the femoral or radial artery.

#350 · Clinical Principles Procedures and Contrast Pharmaceuticals

Name three therapeutic interventional radiology procedures.

Answer

Angioplasty (balloon dilation of a stenotic vessel), stent placement, embolization, or thrombolysis.

#351 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is the physical principle behind DEXA bone density measurement?

Answer

Differential attenuation of two distinct X-ray energy levels through bone and soft tissue allows the soft tissue component to be mathematically subtracted, isolating attenuation attributable to bone mineral content.

#352 · Clinical Principles Procedures and Contrast Pharmaceuticals

What T-score range defines osteopenia on a DEXA scan?

Answer

A T-score between negative 1.0 and negative 2.5.

#353 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is the difference between a T-score and a Z-score on a DEXA report?

Answer

The T-score compares the patient to peak bone density of a healthy young adult of the same sex; the Z-score compares the patient to age-, sex-, and sometimes ethnicity-matched peers.

#354 · Clinical Principles Procedures and Contrast Pharmaceuticals

What is the difference between the craniocaudal (CC) and mediolateral oblique (MLO) mammographic views?

Answer

The CC view passes vertically superior to inferior through the compressed breast, capturing the medial and central portions well, while the MLO view is angled to capture the greatest volume of breast tissue, including the axillary tail and upper outer quadrant.

#355 · Clinical Principles Procedures and Contrast Pharmaceuticals

What does a BI-RADS category 4 mammographic finding indicate?

Answer

A suspicious finding warranting biopsy, further subdivided into 4A, 4B, and 4C to reflect increasing degrees of suspicion.

#356 · Image and Data Quality Analysis

What three things does a technologist decide when critiquing a finished radiograph?

Answer

Whether the image is diagnostic, whether there is a technical or positioning flaw that must be corrected, and whether the image needs to be repeated.

#357 · Image and Data Quality Analysis

Why is image critique considered a patient safety skill rather than an aesthetic judgment?

Answer

Because technical or positioning flaws can hide clinically important findings, such as a pneumothorax at the lung apex or a subtle fracture line, directly affecting patient care.

#358 · Image and Data Quality Analysis

When is the single most effective quality control action a technologist can take on every examination?

Answer

When a systematic image critique is performed immediately at the console, before the patient leaves the department.

#359 · Image and Data Quality Analysis

List the six elements of the systematic image critique sequence in the order commonly taught.

Answer

Positioning, exposure and density, contrast, sharpness and detail, artifacts, and markers.

#360 · Image and Data Quality Analysis

Fill in the blank: A technically perfect exposure of an incorrectly positioned patient is still a ____ image.

Answer

non-diagnostic

#361 · Image and Data Quality Analysis

Why is positioning accuracy evaluated first in the critique sequence?

Answer

Because even a technically perfect exposure is non-diagnostic if the patient was positioned incorrectly.

#362 · Image and Data Quality Analysis

What four questions does a technologist ask when evaluating positioning accuracy?

Answer

Are the correct landmarks visible and symmetrically demonstrated, is the correct amount of anatomy included within the borders, is any rotation present, and does the projection show what the requisition asked for.

#363 · Image and Data Quality Analysis

On a PA chest projection, what confirms the patient was not rotated?

Answer

The medial ends of the clavicles are equidistant from the spinous processes.

#364 · Image and Data Quality Analysis

On a PA chest radiograph, what must be included within the field of view to confirm adequate positioning?

Answer

Both costophrenic angles.

#365 · Image and Data Quality Analysis

Scenario: A lateral cervical spine radiograph does not show the C7-T1 junction. What is the most likely positioning error?

Answer

Failure to depress the shoulders enough to demonstrate the lower cervical spine.

#366 · Image and Data Quality Analysis

How many vertebrae, plus the level below, should be visible on a lateral cervical spine projection?

Answer

All seven cervical vertebrae and the C7-T1 junction.

#367 · Image and Data Quality Analysis

What controls the overall degree of blackening (density) on a radiographic image?

Answer

Milliamperage-seconds (mAs), primarily.

#368 · Image and Data Quality Analysis

What does the numeric exposure indicator on a digital receptor tell the technologist?

Answer

Whether the receptor actually received an appropriate quantity of radiation, regardless of how the final displayed brightness was adjusted by post-processing.

#369 · Image and Data Quality Analysis

Fill in the blank: Digital image processing can partially _____ an exposure error by automatically adjusting the displayed brightness and contrast.

Answer

mask

#370 · Image and Data Quality Analysis

Why should a technologist always check the numeric exposure indicator rather than relying on visual appearance alone?

Answer

Because a digitally processed image can look acceptable on screen even when the receptor dose was too low or too high.

#371 · Image and Data Quality Analysis

What controls contrast on a radiographic image?

Answer

Kilovoltage, the use of grids, and digital post-processing algorithms.

#372 · Image and Data Quality Analysis

What does adequate contrast allow the eye to do?

Answer

Distinguish between adjacent tissues of similar density, such as soft tissue planes or a subtle lung nodule from surrounding lung parenchyma.

#373 · Image and Data Quality Analysis

Describe the appearance of an image with excessive contrast.

Answer

Very black and very white areas with little grey scale in between, which can obscure detail in tissues of intermediate density.

#374 · Image and Data Quality Analysis

Describe the appearance of an image with insufficient contrast.

Answer

A flat, grey image in which structures of differing density blend together and fine detail is lost.

#375 · Image and Data Quality Analysis

What geometric factors affect the sharpness of recorded detail?

Answer

Focal spot size, source-to-image distance, and object-to-image distance.

#376 · Image and Data Quality Analysis

What is the most common and most preventable cause of unsharpness in clinical practice?

Answer

Patient or equipment motion during the exposure.

#377 · Image and Data Quality Analysis

Define artifact in the context of radiographic imaging.

Answer

Any unwanted density or pattern on the image that does not represent true anatomy.

#378 · Image and Data Quality Analysis

Why is the anatomical side marker check given special weight in both clinical practice and on the certification exam?

Answer

Because an incorrect or missing marker is treated as a critical error regardless of how well every other element of the image was performed.

#379 · Image and Data Quality Analysis

What must be true about the sequence in which the critique elements are checked?

Answer

A technologist may reorder the steps to suit personal workflow, but every element must be checked before the patient leaves the room.

#380 · Image and Data Quality Analysis

On chest radiography, what structure should be seen in the midline?

Answer

The trachea.

#381 • Image and Data Quality Analysis

Where is the gastric air bubble normally located on a chest radiograph?

Answer

Beneath the left hemidiaphragm.

#382 • Image and Data Quality Analysis

How many posterior ribs should be visible above the diaphragm on a properly inspired chest radiograph?

Answer

Ten posterior ribs.

#383 · Image and Data Quality Analysis

What landmarks does a technologist check on abdominal radiography for positioning and for recognizing pathology such as free air or obstruction?

Answer

The psoas muscle shadows, renal outlines when visible, the properitoneal fat stripe, and the gas pattern within bowel loops.

#384 · Image and Data Quality Analysis

On extremity radiography, what confirms a true lateral knee projection?

Answer

Superimposition of the femoral condyles.

#385 • Image and Data Quality Analysis

What relationship confirms adequate internal rotation on an ankle mortise view?

Answer

The specific relationship of the distal fibula (lateral malleolus) to the tibia and talus.

#386 • Image and Data Quality Analysis

What landmarks are used on extremity radiography besides joint spaces and cortical margins?

Answer

Epiphyseal plates in the pediatric patient and the specific bony landmarks used to confirm a true anteroposterior or lateral projection.

#387 · Image and Data Quality Analysis

What reference points are used on skull and facial bone imaging?

Answer

The orbital rims, the petrous ridges, the mandibular condyles, and the maxillary sinuses.

#388 · Image and Data Quality Analysis

What structures does a technologist check for symmetry, alignment, and adequate penetration on spine imaging?

Answer

The pedicles, spinous processes, intervertebral disc spaces, and the vertebral body endplates.

#389 · Image and Data Quality Analysis

Why is recognizing normal anatomical landmarks not simply an anatomy exercise?

Answer

It is the mechanism by which a technologist confirms a projection is correctly performed, and it is the same vocabulary a radiologist uses when describing a finding.

#390 · Image and Data Quality Analysis

Fill in the blank: An artifact is any density, marking, or pattern on a radiographic image that does not represent actual patient ____.

Answer

anatomy

#391 · Image and Data Quality Analysis

What does motion artifact look like on a radiograph?

Answer

Blurring or streaking of anatomical margins.

#392 · Image and Data Quality Analysis

List three causes of motion artifact.

Answer

Voluntary patient movement, involuntary movement such as peristalsis or cardiac pulsation, and breathing during the exposure.

#393 · Image and Data Quality Analysis

What corrective actions reduce motion unsharpness?

Answer

Using the shortest exposure time practical, clear and well-timed patient communication and breathing instructions, immobilization devices, and suspending exposure until involuntary tremor has settled where possible.

#394 · Image and Data Quality Analysis

What does grid cutoff typically produce on an image?

Answer

Uneven density, most commonly seen as reduced density at the periphery of the image or across the entire image.

#395 · Image and Data Quality Analysis

What causes off-level grid cutoff?

Answer

The tube and grid are not parallel.

#396 · Image and Data Quality Analysis

What causes off-centre grid cutoff?

Answer

The central ray is not aligned with the centre of the grid.

#397 · Image and Data Quality Analysis

What causes off-focus grid cutoff?

Answer

The source-to-image distance falls outside the grid's designed focal range.

#398 · Image and Data Quality Analysis

What grid error is produced by upside-down focused grid placement?

Answer

Grid cutoff, since a focused grid must be positioned correctly for its lead strips to align with the diverging x-ray beam.

#399 • Image and Data Quality Analysis

Why does recognizing the specific pattern of density loss from grid cutoff matter?

Answer

It helps the technologist identify which type of grid error occurred so it can be corrected before repeating the exposure.

#400 • Image and Data Quality Analysis

Scenario: An image shows symmetric bilateral cutoff versus cutoff concentrated on one side. What does comparing these patterns help identify?

Answer

Which type of grid alignment error occurred, for example off-level versus off-centre.

#401 · Image and Data Quality Analysis

What causes double exposure on a digital radiography system?

Answer

Two separate exposures made on the same image receptor without an intervening readout.

#402 · Image and Data Quality Analysis

How is double exposure prevented?

Answer

By confirming the receptor has been cleared or by using correct workflow steps in the digital radiography system to select a new image for each exposure.

#403 · Image and Data Quality Analysis

What CR/DR handling errors produce scratches, plate fog, and processing artifacts?

Answer

Improper computed radiography and digital radiography plate or detector handling.

#404 · Image and Data Quality Analysis

What causes plate fog on a computed radiography cassette?

Answer

Scattered radiation exposure during storage.

#405 · Image and Data Quality Analysis

How are CR/DR plate handling artifacts prevented?

Answer

Correct plate handling, storage away from sources of scatter, and routine equipment quality control.

#406 · Image and Data Quality Analysis

Describe the appearance of a static artifact.

Answer

Branching, tree-like, or spotty density patterns.

#407 · Image and Data Quality Analysis

What causes static artifacts?

Answer

Electrostatic discharge, most often related to low humidity environments and friction between the image receptor and its handling surfaces.

#408 · Image and Data Quality Analysis

How is static artifact occurrence reduced?

Answer

Maintaining appropriate humidity levels in the imaging suite and handling cassettes and plates carefully.

#409 · Image and Data Quality Analysis

What is the technologist's responsibility regarding externally worn radiopaque items before exposure?

Answer

To identify and remove removable external objects before exposure whenever clinically appropriate.

#410 · Image and Data Quality Analysis

Scenario: An unexpected radiopaque object is visualized on a finished image. What should the technologist do?

Answer

Recognize that it warrants clinical correlation rather than simply assuming it is an artifact.

#411 · Image and Data Quality Analysis

List examples of foreign object artifacts internal to the patient.

Answer

Surgical clips, retained medical devices, or swallowed items.

#412 · Image and Data Quality Analysis

What causes a digital processing artifact where the image appears markedly too light or too dark despite appropriate exposure?

Answer

Improper histogram analysis leading to inappropriate automatic brightness and contrast scaling, particularly when the collimated field or anatomy does not match what the system's algorithm expects.

#413 · Image and Data Quality Analysis

Fill in the blank: A _____ artifact appears as scratches, plate fog, or processing marks from improper CR or DR handling.

Answer

handling

#414 · Image and Data Quality Analysis

What clinical items can cause external foreign object artifacts?

Answer

Jewellery, snaps, buttons, hairpins, or clothing with radiopaque decoration.

#415 · Image and Data Quality Analysis

What visual appearance does quantum mottle produce on an underexposed digital image?

Answer

A mottled or grainy appearance.

#416 · Image and Data Quality Analysis

What causes quantum mottle on a digital image?

Answer

An insufficient number of x-ray photons reaching the receptor to form a smooth image.

#417 · Image and Data Quality Analysis

What is the correction for an underexposed digital image?

Answer

Increase milliamperage-seconds, since mAs is the primary controller of the quantity of radiation delivered and therefore of receptor dose and image noise.

#418 · Image and Data Quality Analysis

Why does simply brightening a displayed underexposed image through post-processing not correct the problem?

Answer

It does not correct the underlying noise problem and does not change the fact that receptor dose was inadequate.

#419 · Image and Data Quality Analysis

Scenario: An image's exposure indicator is above the acceptable range but the image appears visually acceptable on screen. What does this represent?

Answer

Overexposure, which digital receptors' wide dynamic range and automatic processing can mask visually, which is why the numeric exposure indicator must always be checked.

#420 · Image and Data Quality Analysis

What is the correction for an overexposed digital image?

Answer

Reduce milliamperage-seconds for the next exposure and document or flag the occurrence per departmental quality assurance protocol.

#421 · Image and Data Quality Analysis

Why is overexposure a concern even when the resulting image looks acceptable?

Answer

It represents unnecessary patient radiation dose.

#422 · Image and Data Quality Analysis

Why does repeated overexposure across multiple exams indicate a deeper problem?

Answer

It indicates a technique chart or workflow problem that should be corrected at the source.

#423 · Image and Data Quality Analysis

What is the primary correction for poor image contrast?

Answer

Kilovoltage adjustment and, where used, appropriate grid selection.

#424 · Image and Data Quality Analysis

How does kilovoltage relate to contrast?

Answer

Kilovoltage controls the penetrating power of the beam and therefore the scale of contrast produced in the image.

#425 · Image and Data Quality Analysis

How is scatter radiation, which degrades contrast, controlled?

Answer

Through collimation and grid use.

#426 · Image and Data Quality Analysis

Can window width and window level adjustments at the review workstation substitute for correct technical factor selection at the time of exposure?

Answer

No. They can improve displayed contrast for interpretation, but they do not substitute for correct technical factors with respect to patient dose and image noise fixed at the moment of exposure.

#427 · Image and Data Quality Analysis

How does motion blur differ in pattern from geometric unsharpness?

Answer

Motion blur has a directional quality, often following the direction of movement, while geometric unsharpness tends to affect the entire image or a structure uniformly.

#428 · Image and Data Quality Analysis

Scenario: A chest radiograph shows sharp detail in the upper thorax but blurred diaphragm and lower lung fields. What is the likely cause?

Answer

Motion unsharpness from breathing during the exposure.

#429 · Image and Data Quality Analysis

What causes geometric unsharpness?

Answer

A large focal spot or excessive object-to-image distance.

#430 · Image and Data Quality Analysis

How is motion unsharpness corrected?

Answer

By reducing exposure time and improving patient communication, cooperation, and immobilization.

#431 · Image and Data Quality Analysis

How is geometric unsharpness corrected?

Answer

By focal spot selection and positioning the anatomy of interest as close to the image receptor as clinically possible.

#432 · Image and Data Quality Analysis

Why is correct placement of the anatomical side marker one of the most heavily emphasized quality elements in Canadian radiographic practice?

Answer

Because the consequence of an error is severe and can lead to a surgical or interventional procedure being performed on the wrong side.

#433 · Image and Data Quality Analysis

When must the anatomical side marker be placed?

Answer

Within the collimated field of view at the time of exposure whenever possible.

#434 · Image and Data Quality Analysis

What must the anatomical side marker correctly identify?

Answer

The true anatomical side of the patient being imaged, not the side convenient for the technologist's workflow.

#435 · Image and Data Quality Analysis

Is post-exposure digital annotation of a side marker that was not physically present during the exposure a recommended routine practice?

Answer

No, it is discouraged as routine practice and, where policy permits it at all, is reserved for specific documented circumstances.

#436 · Image and Data Quality Analysis

Why does an electronically added marker carry higher risk than a marker physically present during exposure?

Answer

It carries a higher risk of human error and does not provide a verifiable record that the correct side was identified and confirmed at the time the patient was on the table.

#437 · Image and Data Quality Analysis

What must a technologist verify about patient identification annotation before finalizing an image?

Answer

That it is accurate, whether embedded automatically from the radiology information system or entered manually.

#438 · Image and Data Quality Analysis

Why is an image associated with the wrong patient record a serious error?

Answer

It can lead to a diagnosis being attributed to the wrong individual or a needed examination being missed for the correct patient.

#439 · Image and Data Quality Analysis

Scenario: A technologist discovers a marker error before the patient has left the department, and there is doubt about which side was actually imaged. What is the safest response?

Answer

Repeat the exposure with the correct marker physically present in the field at the time of the new exposure.

#440 · Image and Data Quality Analysis

Under what limited condition may a documented correction procedure be used instead of repeating the exposure for a marker error?

Answer

When the correct side can be confirmed with certainty and departmental policy allows a specific documented correction procedure.

#441 • Image and Data Quality Analysis

What should a technologist never do when a marker error is discovered?

Answer

Never guess at which side was imaged after the fact, and never permanently obscure or remove information that could help resolve the ambiguity.

#442 • Image and Data Quality Analysis

What must happen when a marker error is discovered after the patient has left the department?

Answer

The event must be documented according to the facility's incident reporting or quality assurance protocol.

#443 • Image and Data Quality Analysis

What is the purpose of documenting marker errors and repeated examinations?

Answer

To support patient safety and departmental quality improvement by allowing trends to be identified and addressed through education or workflow change.

#444 • Image and Data Quality Analysis

Fill in the blank: A professional culture of transparent incident reporting, rather than _____ of errors, is a foundational expectation of safe radiographic practice in Canada.

Answer

concealment

#445 • Image and Data Quality Analysis

Is incident documentation for a marker error meant to be a punitive record?

Answer

No, it is never treated as a punitive record whose purpose is to penalize an individual technologist for an honest error.

#446 • Image and Data Quality Analysis

What pattern on a chest radiograph may represent pneumonia, atelectasis, or consolidation?

Answer

A wedge-shaped area of increased density within a lung field.

#447 · Image and Data Quality Analysis

Scenario: A technologist notices a wedge-shaped density on a chest image. What should the technologist confirm about the image?

Answer

That it adequately demonstrates the full extent of the finding, including frontal and lateral projections where the requisition and protocol call for them.

#448 · Image and Data Quality Analysis

What pattern on a chest radiograph may represent a pneumothorax?

Answer

A thin line of lucency paralleling the chest wall, separated from lung markings.

#449 · Image and Data Quality Analysis

Why might an expiratory chest image obscure a small pneumothorax?

Answer

Because an expiratory image can hide a small pneumothorax that would be visible on a properly inspired image.

#450 · Image and Data Quality Analysis

What does a sharp lucent line crossing normal bony cortex most likely represent?

Answer

A probable fracture.

#451 · Image and Data Quality Analysis

Scenario: A wrist series demonstrates a fracture but only includes one projection of the forearm. What is the concern?

Answer

An incomplete series can miss a second, less obvious fracture elsewhere along the same bone, so both standard orthogonal projections and adjacent joints should be demonstrated where clinically indicated.

#452 · Image and Data Quality Analysis

What does free air beneath the diaphragm on an upright or lateral decubitus abdominal image represent?

Answer

A surgical emergency indicating perforation of a hollow viscus.

#453 · Image and Data Quality Analysis

Scenario: A technologist recognizes free air under the diaphragm on an abdominal image. What is the appropriate response?

Answer

Understand the urgency and promptly make the image available for interpretation rather than treating the examination as routine.

#454 · Image and Data Quality Analysis

When a technologist observes an unexpected or urgent finding, what is the appropriate professional response?

Answer

Prompt communication through the appropriate channel, typically notifying the radiologist or ordering physician's team according to departmental critical results protocol.

#455 · Image and Data Quality Analysis

Should a technologist verbally communicate a diagnosis to a patient after noticing an abnormal finding?

Answer

No, diagnosis and disclosure of results remain within the scope of practice of the interpreting physician.

#456 · Image and Data Quality Analysis

What is the technologist's role when an urgent finding is observed?

Answer

Ensure image quality is adequate to demonstrate the finding, escalate through appropriate clinical channels, and document the observation according to departmental policy.

#457 • Image and Data Quality Analysis

Scenario: An image is rotated and underexposed but also shows an obvious abnormal finding. Should it be repeated?

Answer

Yes, an obvious finding does not excuse technical deficiencies elsewhere in the image, so the image still requires a repeat for the technical reasons.

#458 • Image and Data Quality Analysis

Scenario: An image meets all technical and positioning criteria but shows an unexpected concerning finding. Should it be repeated?

Answer

No, a technologist should never repeat an examination purely because a finding is present if the image itself meets all technical and positioning criteria.

#459 · Image and Data Quality Analysis

Why should a technically adequate diagnostic image never be repeated solely because it shows a concerning finding?

Answer

Repeating it exposes the patient to additional unnecessary radiation dose and delays the interpretation of an image that already demonstrates what is clinically needed.

#460 · Image and Data Quality Analysis

What is the difference between a technologist noticing pathology and diagnosing it?

Answer

A technologist's trained eye may be first to notice an abnormality, but making the diagnosis remains the responsibility of the interpreting radiologist or physician.

#461 • Image and Data Quality Analysis

What benefit does consistently applying the systematic critique sequence provide to a department?

Answer

It prevents the majority of repeat examinations, protects patients from unnecessary additional radiation dose, and produces consistently diagnostic images.

#462 • Image and Data Quality Analysis

What is a departmental peer review program?

Answer

A program in which technologists review a sample of each other's completed examinations against an objective critique standard.

#463 • Image and Data Quality Analysis

What systemic issues can a peer review program identify?

Answer

A technique chart entry that consistently underexposes a certain body habitus, or a recurring positioning error on a specific projection indicating a need for staff education.

#464 • Image and Data Quality Analysis

What kind of exam questions should candidates expect from the image and data quality analysis domain?

Answer

Scenario-based questions presenting a described or depicted image with a specific flaw, asking the candidate to identify the flaw's cause, the appropriate corrective technical factor change, or the appropriate professional response.

#465 • Image and Data Quality Analysis

Why does correcting a technical error after the patient has left the department carry extra cost?

Answer

It means calling the patient back, which delays diagnosis, increases the patient's radiation dose from a second examination, and wastes departmental resources.

#466 • Image and Data Quality Analysis

What quality review mechanisms exist for radiographs in a Canadian imaging department?

Answer

An informal check by the technologist at the console, a formal peer review program, and an accreditation body's periodic image audit.

#467 · Image and Data Quality Analysis

Fill in the blank: The technologist's own critique at the time of image acquisition is the first and most important checkpoint because it happens _____ the patient has left the department.

Answer

before

#468 · Image and Data Quality Analysis

Scenario: A knee radiograph shows the femoral condyles clearly separated rather than superimposed. What positioning error does this suggest?

Answer

The knee was not truly lateral; rotation is present.

#469 · Image and Data Quality Analysis

What does symmetric demonstration of anatomy on a projection typically confirm?

Answer

That no rotation is present distorting the true anatomical relationships.

#470 · Image and Data Quality Analysis

What is checked to confirm the correct amount of anatomy is included in a projection?

Answer

That the anatomy demonstrated is included within the borders of the image receptor as required by the projection's positioning criteria.

#471 • Image and Data Quality Analysis

Why must the projection actually show what the requisition asked for?

Answer

Because an image that is technically well produced but does not answer the clinical question is still not diagnostic for the ordered examination.

#472 • Image and Data Quality Analysis

What is the relationship between milliamperage-seconds and image noise on a digital system?

Answer

Milliamperage-seconds controls the quantity of radiation delivered to the receptor, and insufficient mAs produces increased image noise, seen as quantum mottle.

#473 · Image and Data Quality Analysis

Scenario: An exposure indicator reads within range, but the displayed image appears too dark. What is the likely explanation?

Answer

A post-processing or windowing issue rather than an actual exposure error, since the numeric exposure indicator reflects the receptor dose independent of display brightness.

#474 · Image and Data Quality Analysis

What visual clue distinguishes excessive contrast from insufficient contrast on a radiograph?

Answer

Excessive contrast shows very black and very white areas with minimal grey scale, while insufficient contrast shows a flat, grey image with structures blending together.

#475 · Image and Data Quality Analysis

Why can differentiating soft tissue planes require adequate contrast rather than just adequate density?

Answer

Because contrast, not density, governs the eye's ability to distinguish between adjacent tissues of similar density.

#476 · Image and Data Quality Analysis

What geometric factor, besides focal spot size, affects sharpness through the distance between the object and the receptor?

Answer

Object-to-image distance.

#477 · Image and Data Quality Analysis

Scenario: An extremity radiograph shows uniform blur across the entire image with no directional streaking. Is this more consistent with motion or geometric unsharpness?

Answer

Geometric unsharpness, since it tends to affect the entire image or a specific structure uniformly rather than showing directional streaking.

#478 · Image and Data Quality Analysis

What is the corrective action focus for geometric unsharpness versus motion unsharpness?

Answer

Geometric unsharpness is corrected through focal spot selection and reducing object-to-image distance; motion unsharpness is corrected through shorter exposure time and better patient cooperation.

#479 · Image and Data Quality Analysis

Scenario: A cassette-based image shows spotty, tree-like density marks after a very dry winter day in the imaging suite. What artifact is this and why?

Answer

A static artifact, caused by electrostatic discharge related to low humidity and friction between the image receptor and handling surfaces.

#480 · Image and Data Quality Analysis

Scenario: An image shows two overlapping sets of anatomy superimposed on one receptor. What artifact is this?

Answer

Double exposure.

#481 · Image and Data Quality Analysis

Scenario: A digital chest image displays markedly too dark despite normal technical factors having been used, and the collimated field is unusually small. What is the likely cause?

Answer

A processing artifact from improper histogram analysis, where the small collimated field does not match what the automatic brightness and contrast algorithm expects.

#482 · Image and Data Quality Analysis

What term describes the appearance of a well-formed digital image with an adequate number of x-ray photons reaching the receptor?

Answer

A smooth image, without the mottled or grainy appearance of quantum mottle.

#483 • Image and Data Quality Analysis

Why is checking the exposure indicator described as more reliable than judging exposure by appearance?

Answer

Because digital processing and a wide dynamic range can make both underexposed and overexposed images look visually acceptable on the display.

#484 • Image and Data Quality Analysis

What professional value underlies documenting a marker error rather than concealing it?

Answer

Transparent incident reporting supports patient safety and quality improvement rather than penalizing the individual technologist.

#485 • Image and Data Quality Analysis

Scenario: A technologist is uncertain whether a right-side marker or a left-side marker was actually present during exposure, and the patient has already left. What should be documented?

Answer

The marker discrepancy must be documented through the facility's incident reporting or quality assurance protocol.

#486 • Image and Data Quality Analysis

What is the shared vocabulary between a technologist evaluating positioning and a radiologist describing a finding?

Answer

Normal anatomical landmarks, such as costophrenic angles, psoas shadows, or vertebral endplates.

#487 · Image and Data Quality Analysis

Fill in the blank: On abdominal radiography, the ____ stripe is a landmark used for positioning and for recognizing pathology such as free air or obstruction.

Answer

properitoneal fat

#488 · Image and Data Quality Analysis

What is the clinical significance of recognizing a pneumothorax pattern early during image critique?

Answer

It prompts the technologist to confirm positioning and inspiration were adequate to demonstrate the finding clearly before the patient leaves.

#489 · Image and Data Quality Analysis

Scenario: An abdominal image is technically excellent but a wedge-shaped chest finding was never asked for on the requisition. Does the technologist need to add extra projections?

Answer

No, the technologist follows the requisition and departmental protocol; additional projections are only added where the requisition and protocol call for them.

#490 · Image and Data Quality Analysis

What is quantum mottle sometimes also described as on a digital image?

Answer

A visually mottled or grainy appearance caused by insufficient photons reaching the receptor.

#491 · Image and Data Quality Analysis

Why should a technologist never permanently obscure or remove information after a marker discrepancy is found?

Answer

Because doing so could prevent resolving the ambiguity about which side was actually imaged.

#492 · Image and Data Quality Analysis

What professional principle governs whether to repeat an exposure after discovering an unexpected finding versus a technical flaw?

Answer

Repeat only for genuine technical or positioning deficiencies; never repeat solely because a finding is present when the image already meets all technical and positioning criteria.

#493 · Image and Data Quality Analysis

Scenario: A lumbar spine image shows adequate penetration but the pedicles are not symmetric bilaterally. What does this suggest?

Answer

Patient rotation, since symmetry of the pedicles is one of the criteria checked for alignment on spine imaging.

#494 · Image and Data Quality Analysis

What is the technologist's escalation duty distinct from making a diagnosis?

Answer

To promptly communicate an unexpected or urgent finding through appropriate clinical channels, such as notifying the radiologist or the ordering physician's team.

#495 · Image and Data Quality Analysis

Fill in the blank: Grid cutoff can occur when the source-to-image distance falls outside the grid's designed _____ range.

Answer

focal

#496 · Image and Data Quality Analysis

What is the key difference between correcting an overexposed image and correcting an underexposed image?

Answer

An underexposed image is corrected by increasing mAs, while an overexposed image is corrected by decreasing mAs for the next exposure.

#497 · Image and Data Quality Analysis

Scenario: A repeat log shows the same projection is consistently overexposed across multiple patients. What does this indicate?

Answer

A technique chart or workflow problem that should be corrected at the source, per departmental quality assurance protocol.

#498 · Image and Data Quality Analysis

What determines whether a technologist should reorder the standard critique sequence?

Answer

Personal workflow preference is acceptable as long as every element of the sequence is checked before the patient leaves the room.

#499 · Image and Data Quality Analysis

Why is a missing anatomical side marker treated as a critical error even on an otherwise perfect image?

Answer

Because the marker's accuracy directly affects patient safety, including the risk of a wrong-side surgical or interventional procedure.

#500 · Image and Data Quality Analysis

What overall skill does the image and data quality analysis domain test on the CAMRT exam?

Answer

The integration of technical knowledge, anatomical knowledge, and professional judgment needed to critique a finished image and decide whether it is diagnostic, flawed, or requires a repeat.